

PLP Architecture  
**60 Curzon Street, London**  
Stage 4 Façade Specification

Issue 4.2 rev 2 | 10<sup>th</sup> January 2019

Job number 226066-00

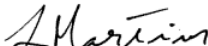
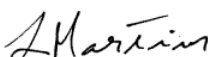
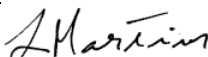
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# Document Verification

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|                       |                  | <b>Name</b>                                                                         | Darren Kent                                                                                      | Darren Kent                                                                           | Darren Kent                                                                         |
|                       | <b>Signature</b> |  |               |  |                                                                                     |

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# 1 Introduction

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60 Curzon Street is located in London's Mayfair district on the corner of Curzon and Bolton Streets. It comprises of the demolition of the existing outdated 1930's apartment block and the construction of a new high end luxury apartment building.

The building consists of nine storeys above the ground floor and a further five floors of accommodation below the ground floor. The building contains 32 apartments, ranging in size from one bedroom to five bedrooms, a ground floor restaurant, Spa and wellness facility, private car parking garage and landscaped gardens to the rear.

## 1.1 The Works

The Contract Documents, including the Architects Drawings and this Specification, define the façade elements of the Works, the performance criteria, geometry of the exposed surfaces, joint positions and detail the required materials and visible finishes. The façade elements of the Works includes a Contractor Design Portion (CDP); as such the Contractor and/or Sub-Contractor(s) shall be responsible for developing and completing the detailed design in accordance with this document and the other Contract Documents.

## 2 Cladding types

---

The façade elements of the Works include the following cladding types:

|                  |                                                    |
|------------------|----------------------------------------------------|
| Cladding Type 1  | Typical windows and doors.                         |
| Cladding Type 2  | Rooftop pavilion - curtain wall windows and doors. |
| Cladding Type 3  | Glazing to core area                               |
| Cladding Type 4  | Glazing and canopies to garden unit entrances      |
| Cladding Type 5  | Glazing, canopies and cladding to main entrance    |
| Cladding Type 6  | Glazed rooflight (Wintergarden)                    |
| Cladding Type 7  | Doors                                              |
| Cladding Type 8  | Balustrades                                        |
| Cladding Type 9  | Metal soffit panels                                |
| Cladding Type 10 | Stone cladding                                     |
| Cladding Type 11 | Rainscreen terracotta tile cladding                |
| Cladding Type 12 | Rendered wall                                      |

Each cladding type and the associated sub-types are described in the following sections of this Specification.

## 2.1 Cladding Type 1.x – typical windows and doors

### 2.1.1 Cladding sub-types

Cladding Type 1 includes window and door sub-types with various sizes and configurations of fixed/openable glazing and flat/curved plan geometry:

|          |                                      |
|----------|--------------------------------------|
| Type 1.1 | Single opening (not curved)          |
| Type 1.2 | Single opening (curved)              |
| Type 1.3 | Double fixed (not curved)            |
| Type 1.4 | Double opening (not curved)          |
| Type 1.5 | Hybrid, fixed & opening (not curved) |
| Type 1.6 | Hybrid, opening (not curved)         |

Refer to glazing/window types matrix for window tag codes; drawing 1137-A-611-xx.

### 2.1.2 System (typical)

- Thermally-broken aluminium frames based upon the Schueco AWS-90 BS system.
- Bespoke extrusions for external capping, head cornice, grille/features.
- Fabricated aluminium cills and fascia panels (3mm thick);
- Bronze coloured anodised aluminium finishes;
- Patinated stainless steel closure trims at perimeter interface between windows and stone.
- Openable elements are side hung and open inwards with restrictor stays concealed within the framing.
- Double opening elements have no fixed intermediate mullion.
- Window and door frames supported at the base and restrained at the head and jambs by continuous steel angles fixed to concrete primary structure;
- Thermally broken and insulated infill framing incorporating deflection head details;
- Insulating glass units (triple glazed);
- “Bronze”/dark brown coloured gaskets;
- Stainless steel fixings (“pig nose” security type for all visible external fixings);
- Bronze coloured ironmongery. Refer to Architect’s schedule;

### 2.1.3 System (non-typical)

- Bespoke “flush” threshold detail at the base of openable elements to accessible balconies and terrace areas. The “flush” threshold shall be positively drained with vertical height changes limited to a dimension of 20mm (including the compressible gasket). Locations incorporating bespoke “flush” thresholds are denoted with a “B” suffix in window tag code, e.g. window tag code “xx-x-B”.
- Extruded aluminium louvre units. Locations incorporating aluminium louvres in lieu of glazing are denoted with an “L” suffix in window tag code, e.g. window tag code “xx-x-L”.
- Insulating glass units (double glazed). Locations incorporating double glazed units in lieu of triple glazed units are identified on the Window and Opening types Matrix; drawing 1137-A-611-04.
- Insulated louver doors, with outward opening frame profiles and thermally insulated panels; drawings in the series 60CS-PLP-A-404-33.

### 2.1.4 Insulating glass units

Triple glazing shall be as follows:

- Laminated low-iron safety glass outer pane (Class 2 BS EN 12600), with low-e coating to surface #4 and opaque black perimeter frit to surface #1 covering the width of the edge spacer;
- 90% Argon filled cavity;
- Monolithic low-iron glass;
- 90% Argon filled cavity;
- Laminated low-iron safety glass inner pane (Class 1, BS EN 12600), with low-e coating to surface #7;

Double glazing shall be as follows:

- Laminated low-iron safety glass outer pane (Class 2 BS EN 12600), with low-e coating to surface #4 and opaque black perimeter frit to surface #1 covering the width of the edge spacer;
- 90% Argon filled cavity;
- Laminated low-iron safety glass inner pane (Class 1, BS EN 12600), with low-e coating to surface #7;
- All glazing to have bronze coloured painted stainless steel spacers to achieve required thermal performance.

### 2.1.5 Details

- The typical ground floor windows on Curzon Street and Bolton Street include retractable external shading awnings. Product type “Marlesbury Awning” supplied by Morco Awnings or equivalent; subject to approval.
- Includes the completion of the weathering, thermal and vapour control lines at interfaces with the primary structure and adjacent cladding types;
- Includes insect mesh and blanking panels to louvers;
- Includes plenum boxes, connection spigots and blanking plates;
- Includes continuous steel angles around the perimeter to provide acoustic continuity with the concrete support;

### 2.1.6 Interfaces

Cladding Type 1 interfaces with:

- Primary structure;
- Internal blinds / curtains;
- Internal finishes;
- Mechanical ductwork for apartment ventilation;
- Stone, balustrade, external grill support structure;
- Adjacent cladding types

## 2.2 Cladding Type 2.x – rooftop pavilion curtain wall, windows, doors, and roof

### 2.2.1 Cladding sub-types

Cladding Type 2 includes curtain wall, window and door sub-types with various sizes and configurations of fixed/openable glazing.

|          |                           |
|----------|---------------------------|
| Type 2.1 | Single fixed              |
| Type 2.1 | Single opening            |
| Type 2.3 | Double fixed              |
| Type 2.4 | Double opening            |
| Type 2.5 | Hybrid, fixed & opening   |
| Type 2.6 | Clerestory glazing, fixed |
| Type 2.8 | Roof                      |

Refer to glazing/window types matrix for window tag codes; drawing 1137-A-611-xx.

### 2.2.2 System (typical details)

- Thermally-broken aluminium frames based upon the Schueco AWS-90 BS and FW-50 systems.
- Bespoke extrusions for external capping, cornice and features.
- Fabricated aluminium cills and fascia panels (3mm thick);
- Bronze coloured anodised aluminium finishes;
- Patinated stainless steel nosing to vertical cornice;
- Openable elements are side hung and open inwards with restrictor stays concealed within the framing.
- Double opening elements have no fixed intermediate mullion.
- Window and door frames supported at the base and restrained at the head and jambs by continuous steel angles;
- Thermally broken and insulated infill framing;
- Insulating glass units (triple glazed);
- Bronze/dark brown coloured gaskets;
- Stainless steel fixings (“pig nose” security type for all visible external fixings);
- Bronze coloured ironmongery. Refer to Architect’s schedule;

### 2.2.3 System (non-typical)

- Bespoke “flush” threshold detail at the base of openable elements to accessible terrace areas. The “flush” threshold shall be positively drained with vertical height changes limited to a dimension of 20mm (including the compressible gasket). Locations incorporating bespoke “flush” thresholds are denoted with a “B” suffix in window tag code, e.g. window tag code “xx-x-B”.
- Insulating glass units (double glazed). Locations incorporating double glazed units in lieu of triple glazed units are identified on the Window and Opening types Matrix; drawing 1137-A-611-04.

### 2.2.4 System (roof)

- Bespoke folded hook-on system roof tiles, 3mm minimum thickness patinated stainless steel sheet of the series 1.43xx, finish to the Architect’s specification
- Acoustic inlay on the back of the roof tile system, performance in accordance with the Acoustician report
- Tile system fixing system, “top-hat”
- Continuous weatherproof membrane lapped and sealed onto interfacing construction
- External grade insulation mechanically fixed with non-ferrous fixings and cavity barriers as required by the Fire Engineer
- Continuous vapour barrier, lapped and sealed onto interfacing construction

### 2.2.5 Support (roof)

- Light steel framing system supporting the roof elements, spanning between the primary steel structure

### 2.2.6 Insulating glass units

Triple glazing shall be as follows:

- Laminated low-iron safety glass outer pane (Class 2 BS EN 12600), with low-e coating to surface #4 and opaque black perimeter frit to surface #1 covering the width of the edge spacer;
- 90% Argon filled cavity;
- Monolithic low-iron glass;
- 90% Argon filled cavity;
- Laminated low-iron safety glass inner pane (Class 1, BS EN 12600), with low-e coating to surface #7;

Double glazing shall be as follows:

- Laminated low-iron safety glass outer pane (Class 2 BS EN 12600), with low-e coating to surface #4 and opaque black perimeter frit to surface #1 covering the width of the edge spacer;
- 90% Argon filled cavity;
- Laminated low-iron safety glass inner pane (Class 1, BS EN 12600), with low-e coating to surface #7;

All glazing to have bronze coloured painted stainless steel spacers to achieve required thermal performance.

### 2.2.7 Details

- Includes the completion of the weathering, thermal and vapour control lines at interfaces with the primary structure and adjacent cladding types;
- Includes insect mesh and blanking panels to louvers;
- Includes plenum boxes, connection spigots and blanking plates;
- Includes continuous steel angles around the perimeter to provide acoustic continuity with the concrete support;

### 2.2.8 Interfaces

Cladding Type 2 interfaces with:

- Primary structure;
- Internal blinds / curtains;
- Internal finishes;
- Terrace finishes;
- Adjacent cladding types;

## 2.3 Cladding Type 3.x – glazing to core area

### 2.3.1 Cladding sub-types

Refer to glazing/window types matrix for window tag codes; drawing 1137-A-611-xx.

### 2.3.2 System (typical)

- Thermally-broken aluminium framing based upon the Schueco FW-50 system;
- Bespoke extrusions for external capping;
- Aluminium cills, panels and closure sheets (3mm thick);
- Bronze coloured anodised aluminium finishes;
- Framing supported floor by floor;
- Insulating glass units (double glazed);
- Bronze/dark brown coloured gaskets;
- Stainless steel fixings (“pig nose” security type for all visible external fixings);

### 2.3.3 Insulating glass units

Double glazing shall be as follows:

- Laminated low-iron safety glass outer pane (Class 2 BS EN 12600), with low-e coating to surface #4 and opaque black perimeter frit to surface #1 covering the width of the edge spacer;
- 90% Argon filled cavity;
- Laminated low-iron safety glass inner pane (Class 1, BS EN 12600), with low-e coating to surface #7;

All glazing to have bronze coloured painted stainless steel spacers to achieve required thermal performance.

### 2.3.4 Details

- Includes the completion of the weathering, thermal and vapour control lines at interfaces with the primary structure and adjacent cladding types;
- Includes continuous steel angles around the perimeter to provide acoustic continuity;

### 2.3.5 Interfaces

Cladding Type 3.1 interfaces with:

- Primary structure;

- Internal finishes;
- Adjacent cladding types;

## 2.4 Cladding Type 4.x – Glazing and canopies to garden unit entrances

### 2.4.1 Cladding sub-types

Glazing system (including side lights), canopies and doors.

Refer to glazing/window types matrix; drawing 1137-A-611-xx illustrating the scope of Cladding Type 4.1.

### 2.4.2 System (typical details)

- Thermally-broken aluminium framing based upon the Schueco FW-50 system;
- Bespoke extrusions for external capping;
- Aluminium cills, panels and closure sheets (3mm thick);
- Bronze coloured anodised aluminium finishes;
- Base supported glazing system;
- Insulating glass units (double glazed);
- Bronze/dark brown coloured gaskets;
- Stainless steel fixings (“pig nose” security type for all visible external fixings);
- Bespoke glazed terracotta tiles;
- Canopy: Laminated glass with bronze anodised aluminium framing elements;
- Secondary structural steel torsion beams and posts providing support to canopy and glazed screen;
- Bronze anodised aluminium ironmongery to Architects schedule;

#### 2.4.2.1 Entrance doors

- Single-leaf glazed doors;
- Bespoke and thermally-broken framing profiles;
- Bronze ironmongery to Architects schedule;
- Bronze/dark brown coloured gaskets;

### 2.4.3 Insulating glass units

- Laminated low-iron glass outer pane (Class 2, BS EN 12600), with low-e coating to surface #4 and opaque black perimeter frit to surface #1 covering the width of the edge spacer;
- 90% filled cavity;
- Laminated low-iron safety glass inner pane (Class 2, BS EN 12600), with low-e coating to surface #7;

All glazing to have bronze coloured painted stainless steel spacers to achieve required thermal performance.

### 2.4.4 Details

- Includes the completion of the weathering, thermal and vapour control lines at interfaces with Ground Floor and adjacent cladding types;

### 2.4.5 Interfaces

Cladding Type 4.1 interfaces with:

- Primary structure;
- Internal finishes;
- Ground Level waterproofing and finishes;
- Adjacent cladding types;
- Security systems (door access control);

## 2.5 Cladding Type 5.x – Glazing, canopies and cladding to main entrance

### 2.5.1 Cladding sub-types

Inner curtain wall glazing, onyx wall and canopies.

- Outer skin: open joint single laminated glass with Onyx panel;
- Inner skin: bespoke' thermally-broken bronze anodised aluminium curtain wall system providing the thermal line of the façade;
- Canopy: Laminated glass soffit with 'stone-effect' digital print and bronze anodised aluminium framing;

Refer to glazing/window types matrix; drawing 1137-A-611-xx illustrating the scope of Cladding Type 5.1.

### 2.5.2 Support

- Onyx laminated skin supported back to the curtain wall system;
- Curtain wall system supported by the main structure at every level;

### 2.5.3 Finish

- Bronze anodised aluminium;
- Onyx stone;

### 2.5.4 Insulating glass units

Triple glazing will be as follows:

- Laminated low-iron safety glass outer pane (Class 2 BS EN 12600), with low-e coating\* to surface #4 and opaque black perimeter frit to surface #1 covering the width of the edge spacer;
- 90% Argon filled cavity;
- Monolithic low-iron glass;
- 90% Argon filled cavity;
- Laminated low-iron safety glass inner pane (Class 1, BS EN 12600), with low-e coating to surface #7;

### 2.5.5 Canopy glass

- Soffit: Laminated glass soffit, incorporating cast glass on the underside and flat heat-strengthened glass above. The glass is to have a 'stone-effect' digital print to surface #2 or surface #3\* and a dichroic coating to surface #3\* or surface #4, the surface is to be confirmed following investigations in

to the compatibility between the coating and the interlayer during the next stage.

\*Note-digital tint and dichroic coating should not be applied on the same glass surface;

### 2.5.6 Glass restraint

- Insulated glass units retained with pressure caps and gaskets. Gaskets to be bronze/dark brown in colour. Final finish to be agreed by the Architect based on approved samples;
- Onyx laminated glass to be locally fixed back to the curtain wall;
- Canopy soffit glass is to be supported by a discrete continuous perimeter bronze frame and intermediate at glass joints. Outer cast glass surface to be recessed at edge on all four sides to allow the outer surface of the glass and support frame to be flush;

### 2.5.7 Details

- Perimeter cladding to canopy: pressed bronze anodised aluminium panels, hooked onto support angle at head and mechanically fixed at base;
- Coping to canopy: Laminated glass structurally bonded to perimeter sub-frame, inclined at 3-5° towards gutter at rear of canopy. Sub-frame is supported by brackets to the canopy structure;
- Includes the completion of the weathering, thermal and vapour control lines at interfaces with Ground Floor and adjacent cladding types;
- Includes bronze anodised external flashings, minimum 3mm thick, to close the gap between the curtain wall and interface details;
- Hollow framing members will be in-filled to satisfy the acoustic flanking transmission and thermal requirements;
- Includes bronze anodised aluminium external flashings, minimum 3mm thick, to close the gap between the canopy, adjacent cladding and the primary structure;
- Includes all canopy support metal work/structure;
- Includes all drainage elements including gutters and downpipes;

### 2.5.8 Interfaces

Cladding Type 5.1 interfaces with:

- Primary structure;
- Internal finishes;
- Adjacent cladding types;
- Security systems (door access control);

## 2.6 Cladding Type 6.x – Glass Rooflight

### 2.6.1 Cladding sub-types

Glass rooflight to Wintergarden.

Refer to glazing/window types matrix; drawing 1137-A-611-xx illustrating the scope of Cladding Type 6.1.

### 2.6.2 System (typical)

- Thermally-broken aluminium framing based upon the Schueco FW-50 system;
- Large format insulated glass panel (double glazed);
- Aluminium perimeter trims and closure sheets (3mm thick);
- Bronze coloured anodised aluminium finishes;
- Insulating glass units (double glazed);
- Bronze/dark brown coloured gaskets;
- Stainless steel fixings (“pig nose” security type for all visible external fixings);
- 3° slope for drainage;
- Glazing face sealed with a low modulus silicone sealant;

### 2.6.3 Insulated glass units

- Heat-soaked tested, thermally toughened low-iron glass outer pane, with low-e coating to surface #2 and opaque black perimeter frit to surface #1 covering the width of the edge spacer;
- 90% Argon filled cavity;
- Laminated low-iron safety glass inner pane (Class 1, BS EN 12600), with low-e coating to surface #3;

### 2.6.4 Details

- Secondary drainage channels that allow any water that permeates the weather seal to drain away;
- Includes the completion of the weatherproofing, insulation and vapour control lines at interfaces adjacent cladding;

### 2.6.5 Interfaces

The glass rooflight interfaces with:

- Primary structure;

- Adjacent cladding types;

## 2.7 Cladding Type 7.x –Doors

### 2.7.1 Cladding sub-types

Refer to door type matrix; drawing 1137-A-611-05.

|          |                                                                   |
|----------|-------------------------------------------------------------------|
| Type 7.1 | Main entrance doors                                               |
| Type 7.2 | Restaurant entrance doors                                         |
| Type 7.3 | Curzon Street Townhouse door                                      |
| Type 7.4 | Bolton Street Ground unit B door and gate                         |
| Type 7.5 | Garage doors                                                      |
| Type 7.6 | Access door to core                                               |
| Type 7.7 | Bolton Street Pavilion roof access door                           |
| Type 7.8 | Bolton Street Lightwell & roof plant access doors (non-insulated) |
| Type 7.9 | Curzon Street Pavilion roof access hatch (non-insulated)          |

Doors are insulated unless otherwise noted.

The doors to the Garden unit entrance are included in cladding type 4.1.

#### 2.7.1.1 Cladding Type 7.1 – Main entrance doors

- Single-leaf solid/opaque doors;
- Decorative bronze feature panels with rectangular relief expression;
- Bespoke and thermally-broken framing profiles;
- Glazed perimeter panels;
- Bronze ironmongery to Architects schedule;
- Bronze/dark brown coloured gaskets;

#### 2.7.1.2 Cladding Type 7.2 – Restaurant entrance doors

- Single-leaf solid/opaque doors;
- Decorative bronze feature door panels with vertical relief expression;
- Bespoke and thermally-broken framing profiles;
- Bronze perimeter framing and portico with insulated curved glass “over-panel” (above door);

- Bronze ironmongery to Architects schedule;
- Bronze/dark brown coloured gaskets;

#### **2.7.1.3 Cladding Type 7.3 – Curzon St. Townhouse Entrance Door**

- Single-leaf solid/opaque door;
- Decorative bronze feature door panels with vertical relief expression;
- Bespoke and thermally-broken framing profiles;
- Insulated glass “over-panel” (above door);
- Bronze ironmongery to Architects schedule;
- Bronze/dark brown coloured gaskets;

#### **2.7.1.4 Cladding Type 7.4 - Ground Unit B Entrance Door**

- Single-leaf solid/opaque door;
- Decorative bronze feature door panels with vertical relief expression;
- Bespoke and thermally-broken framing profiles;
- Bronze ironmongery to Architects schedule;
- Bronze/dark brown coloured gaskets;

#### **2.7.1.5 Cladding Type 7.5 – Garage Doors**

- Custom finished spiral door based upon the SSL Premium “L” version door system supplied by Efaflex Ltd.
- Aluminium frame construction with insulated sandwich infill.
- Decorative bronze fascia panels with horizontal joints/relief to align with adjacent stone rustication.
- Powered opening/closing mechanism – vertical retraction to lower level.

#### **2.7.1.6 Cladding Type 7.6 – Access door to core**

- Single-leaf solid/opaque door;
- Decorative bronze anodised aluminium door panels with vertical fin features ;

- Proprietary thermally-broken framing profiles;
- Bronze ironmongery to Architects schedule;

#### 2.7.1.7 Cladding Type 7.7 – Bolton St. pavilion roof access door

- Single-leaf solid/opaque door;
- Decorative bronze anodised aluminium door panels with vertical fin features ;
- Proprietary thermally-broken framing profiles;
- Bronze ironmongery to Architects schedule;

#### 2.7.1.8 Cladding Type 7.8 – Bolton Street Lightwell & roof plant access doors (non-insulated)

- Single-leaf solid/opaque doors;
- Decorative bronze anodised aluminium door panels with “fluted” vertical relief feature;
- Proprietary framing profiles;
- Bronze ironmongery to Architects schedule;

#### 2.7.1.9 Cladding Type 7.9 – Curzon Street Pavilion roof access hatch (non-insulated)

- Single-leaf solid/opaque doors;
- Bronze anodised aluminium door panels;
- Proprietary framing profiles;
- Bronze ironmongery to Architects schedule;

### 2.7.2 Security & access control requirements

- For security/access control requirements for each door (maglock, access control, keyfob readers. Etc) refer to Architects door schedule and Hoare Lea security layouts.

## 2.8 Cladding Type 8.x – Balustrades

### 2.8.1 Type

- Custom fabricated balustrade assemblies;
- Solid brass handrails (horizontal);
- Bronze vertical profiles;
- Bronze base;
- Countersunk stainless steel “pig nose” bolts at base connection detail;
- Stainless steel sub-structure and brackets;

### 2.8.2 Interfaces

Cladding Type 8 interfaces with:

- Primary structure;
- Adjacent cladding types – windows and stone cladding;

## 2.9 Cladding Type 9.x – Metal Soffit Panels

### 2.9.1 Type

- Bronze anodised aluminium soffit system including the continuity of the thermal line and vapour control layer. Minimum 3mm thick;

### 2.9.2 Support

- Aluminium or stainless steel bracketry fixed to the main structure;

### 2.9.3 Finish

- Bronze anodised aluminium;

### 2.9.4 Details

- Includes bronze anodised aluminium external flashings, minimum 3mm thick, at interface details;
- Includes the completion of the weathering, thermal and vapour control lines at interfaces;
- Includes all secondary metal work supporting soffit;

### 2.9.5 Interfaces

Cladding Type 9 interfaces with:

- Primary structure;
- Cladding Types as per the Architect's drawings;

## 2.10 Cladding Type 10.x – Stone Cladding

### 2.10.1 Type

Stone wall, soffit and coping cladding, typically hand-set unless indicated otherwise.

Includes fluted detail in stone above residential entrance on Curzon Street elevation;

The following types have been identified within this cladding type according to the Architect's design intent:

- 10.1 – Portland Basebed Limestone (honed finish) with flush 6mm joints
- 10.2 – Precast reconstituted stone (to match Portland Basebed Limestone, honed finish) with flush 6mm joints
- 10.3 – White Moleanos (honed finish) with flush 6mm joints
- 10.4 – Beige Moleanos (hand bush hammered finish) with 35x25mm rusticated joints and 6mm joints
- 10.5 – Beige Moleanos (hand bush hammered finish) ) with 45x25mm rusticated joints and 6mm joints
- 10.6 – Beige Moleanos (hand bush hammered finish) with 6mm joints
- 10.7 - Angola Black granite (flamed finish) with 25mm rusticated joints and 6mm joints
- 10.8 – Angola Black granite (flamed finish) with 6mm lime mortar joints
- 10.9 – Terracotta tiles with bespoke glaze with 6mm (vertical) and 10mm (horizontal) joints.

Refer to stone type elevations; drawing series 1137-A-221-xx.

### 2.10.2 Joints in stone

- Typical flush panel to panel joint: 6 mm (nominal) non-staining lime mortar
- Movement joint minimum: 6 mm

All movement joints to be justified by calculations, in accordance with the Movement and Tolerances Report

### 2.10.3 Cladding Zone

- Nominal 75mm thick Portland Basebed Limestone with 6 mm joints filled with lime mortar (except in movement joint locations); stone thickness to be confirmed by calculation.
- Nominal 75mm wide drained and ventilated cavity incorporating stone cladding rail. Main concrete structure tolerances to be accommodated in the cavity. Please refer to the Movement and Tolerance report by the Structural Engineer
- External grade insulation mechanically fixed with non-ferrous fixings and cavity barriers as required by the Fire Engineer.

- Continuous vapour control membrane lapped and sealed onto interfacing construction;
  - “MetSec” type insulated stud frame system substrate filled with mineral wool insulation.
  - Internal finishes;

#### 2.10.4 Support

The support strategy for the stone cladding is as follows:

- Vertical stone elements will be self-supporting with the dead load bearing onto a continuous stainless steel shelf support brackets at the base at slab levels;
- Vertical stone will be restrained at four points, via stainless steel pins located in the top and bed face of each unit. The restraints will be attached to vertical stainless steel sections;
- 2, 3 and 4 storeys high stacking of stone from level 02 on Bolton Street and Curzon Streets;
- 1 storey high stone cladding stacked at every slab level on the rear facades;
- Vertical stainless steel sections will be at approx. 600mm centres;
- Vertical stainless steel sections will be dead load supported at the base and restrained (out-of-plane) to the main structure at every floor level;
- Includes secondary steel support structure to level 7 penthouse terrace (south west elevation).

#### 2.10.5 Brackets

- Grade 1.43xx stainless steel fixed to stainless steel cast-in channels in concrete primary structure;
- Non-metallic thermal isolators as required between brackets and cast-in channels to minimise thermal bridging;

##### 2.10.5.1 Movement joints

- Joints between stacked stone walling elements will be sealed with sealant to allow movement;
- The sizing of the movement joints shall be verified with respect to the sealant flexibility and stone support strategy.

##### 2.10.5.2 Smoke/ fire-stopping

- Non-ferrous cavity barriers will be provided as required by statutory regulations;

##### 2.10.5.3 Details

- Stone will include drips to prevent staining;
- Includes non-ferrous cavity trays over punched openings to drain water in the cavity to outside;

- Includes the completion of the weatherproofing, insulation and vapour control lines at interfaces and with Ground and Roof Level waterproofing and finishes;
- Includes fire stopping and cavity barriers to cavities and around all windows and at compartmentation lines (all slabs and between apartments). Please refer to the Fire Engineers report;
- Includes supporting steelwork where back concrete wall is not present.
- Includes bronze anodised aluminium party wall interface trims and closures.

## 2.10.6 Horizontal and spandrel stone

### 2.10.6.1 Spandrel and Coping composition

- Nominal 75mm thick Portland based limestone spandrel and coping panels (to be confirmed by calculation);
- Cavity;
- External grade insulation mechanically fixed with non-ferrous fixings to the underside of primary structure;
- Vapour control line;

### 2.10.6.2 Soffit composition

- Nominal 75mm thick Portland based limestone soffit panels (to be confirmed by calculation);
- Cavity;
- External grade insulation mechanically fixed with non-ferrous fixings to the underside of primary structure;
- Vapour control line;

### 2.10.6.3 Support

- Panels will be held with a grade 1.43xx stainless steel bracket system;

## 2.10.7 Interfaces

Cladding Type 10 interfaces with:

- Adjacent cladding types;
- Primary structure;
- Ground Level waterproofing and finishes;

## 2.11 Cladding Type 11.x –Terracotta Tiles

### 2.11.1 Type

- Vertical extruded glazed terracotta piers;
- Extruded glazed terracotta surround to level 5-6 terraces;
- Extruded glazed terracotta soffits;
- Slip formed terracotta tiles.

### 2.11.2 Support

- Vertical extruded glazed terracotta piers, each unit will be bottom supported, individually restrained on stainless steel brackets, fixed back to horizontal stainless steel rails that are fixed back to the primary structure;
- Horizontal extruded terracotta elements are individually supported and restrained, fixed back to the window systems;
- Extruded glazed terracotta soffit panels will be held with a grade 1.43xx stainless steel bracket system;

### 2.11.3 Finish

- Extruded glazed terracotta: bespoke ceramic tile finish. Refer to the Architect's drawings and reference samples for further details of colours and proposed glaze by Christine Jetten Studio.
- The bespoke glazed tile finish will be subject to the sample approval procedures described in section;

### 2.11.4 Tile joints

- 6mm vertical
- 10mm horizontal

### 2.11.5 Details

- Includes the completion of the weathering, thermal and vapour control lines at interfaces with Ground Floor and adjacent cladding types;
- Terracotta soffit panels to incorporate a bonded mesh on back face for post breakage integrity of panels;

### 2.11.6 Interfaces

Cladding Type 11 interfaces with:

- Primary structure;

- Adjacent cladding types;

## 2.12 Cladding Type 12 – Rendered wall

### 2.12.1 Type

Proprietary non-ventilated built-up insulated render system on insulated metal stud frame system.

### 2.12.2 Cladding Zone

- Nominal 50mm thick rendered rusticated finish, and exposed natural stone finish, in accordance with the Architect 's Drawings;
- Primer
- Reinforcing coat mineral fibre carrier board, minimum 5mm thick
- Continuous vapour control membrane lapped and sealed onto interfacing construction;
- Vertical rail system in 1.4401 grade stainless steel
- Aluminium starter and edge trims, to suit the manufacturer's recommendations.
- The fixings for the services must not penetrate the vapour barrier.

### 2.12.3 Support

The support strategy for the rendered wall is as follows:

- Metal stud frame spanning from top of slab to underside of slab and is encapsulated with fire resistant mineral fibre carrier board as indicated on the design intent drawings.
- The metal stud frame system shall be designed to accommodate all loads associated with the systems attached to it.
- The metal stud frame system shall also provide sufficient allowance for building movement and tolerances as outlined in the Employer's Requirements.

### 2.12.4 Brackets

- Grade 1.43xx stainless steel fixed to stainless steel cast-in channels in concrete primary structure;
- Non-metallic thermal isolators as required between brackets and cast-in channels to minimise thermal bridging;

#### 2.12.4.1 Movement joints

- Joints between stacked stone walling elements will be sealed with sealant to allow movement;
- The sizing of the movement joints shall be verified with respect to the sealant flexibility and rendered system support strategy.

#### 2.12.4.2 Smoke/ fire-stopping

- Non-ferrous cavity barriers will be provided as required by statutory regulations;

#### 2.12.4.3 Details

- Rendered finish will include drips to prevent staining;
- Includes non-ferrous cavity trays over punched openings to drain water in the cavity to outside;
- Includes the completion of the weatherproofing, insulation and vapour control lines at interfaces and with Garden and Raised Garden waterproofing and finishes;
- Includes fire stopping and cavity barriers to cavities and around all windows and at compartmentation lines (all slabs and between apartments). Please refer to the Fire Engineers report;
- Includes supporting steelwork where back concrete wall is not present.

#### 2.12.5 Interfaces

Cladding Type 12 interfaces with:

- Adjacent cladding types;
- Primary structure;
- Ground Level waterproofing and finishes;

### 3 Prescriptive design requirements

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Set out below are the prescriptive requirements that will be followed in the design of the Works:

- a. All fixings outside or through the vapour control line will be (Grade 4A where visible and grade 2A in all other locations) stainless steel.
- b. All brackets outside the vapour control line will be stainless steel (Grade 1.44xx where visible and grade 1.43xx in all other locations) or aluminium.
- c. Where an inner vapour control line and outer weathering membrane are present in the same wall construction, the vapour control line will have lower water vapour permeability than the outer weathering membrane.
- d. Where flexible or sheet vapour control membranes are connected together or to other systems they will be continuously sealed and positively clamped.
- e. Where insulation must be locally reduced in thickness, high performance insulation will be provided to match the overall performance of the surrounding insulation.
- f. Site applied sealant will not be acceptable as part of the primary sealing system unless shown on the Architect's drawings or noted within this specification.
- g. All gaskets, where possible, will be accessible for inspection/ replacement.
- h. Smoke and fire stops will be provided as required by the Building Regulations.
- i. Monolithic annealed and heat strengthened glass will not be used in "critical locations" as defined in UK Building Regulations Approved Document N.
- j. Toughened glass will not be used unless a technical reason means no other option is available.
- k. The use of heat-soak tested toughened glass will not be used unless a technical reason means no other option is available.
- l. The outer pane of all insulated glass units will be stiffer than the inner most pane to reduce the visual impact of pillowing.
- m. Where glazing is vertical with unsupported edges, horizontal, or sloped (up to 85° degrees from the horizontal) it will be laminated.
- n. Wherever glass performs the role of a barrier, as defined in BS 6180, it will be laminated safety glass Class 1 (BS EN 12600).

- o. Laminated glass balustrades will incorporate at least one pane of either annealed or heat strengthened glass.
- p. Where doors provide access for the disabled, the door threshold upstand will meet the requirements of BS 8300.
- q. Exposed hinges will not be accepted as per the Architect's design intent.
- r. Where manually operated, non-powered swing doors provide access for the disabled, the opening force, when measured at the leading edge of the door, will be in accordance with BS 8300, Clause 6.3.2.
- s. Where the opening force limits noted in o) above mean that the door-closing device is not sufficiently strong to keep the door closed against external conditions, the door closing system will be one of those described in Building Regulations Approved Document M, Clause 2.8.
- t. All powered doors and windows shall be means of concealed drives, i.e. chain driven or spindle mounted. Exposed drive units will not be accepted
- u. The air/ vapour control line will be designed to resist the maximum design wind load. Flexible membranes will not be permitted to span further than 100mm without support.
- v. Glass with un-sealed exposed edges will have polished edges.
- w. Glass forming the edges of an exposed sealant joint will have smooth ground edges.
- x. A non-staining colour matched flexible sealant that is compatible with the stone shall be used to point movement / compression joints.
- y. Mortars used to point will be trialled prior to installation on site to confirm that they do not stain or discolour the stone.
- z. Mitred or butt jointed stone corners will not be accepted unless otherwise agreed by the architect on a case by case basis.
- aa. As per the Architect's requirement all stone to be used in the façade shall be subject to inspection and reviewed by the Architect and Client prior to final cutting;
- bb. As per the Architect's requirement all terracotta tiles shall be reviewed prior to shipping.
- cc. The render system shall be a proprietary system certified with a Agrément Certificate for the proposed design and use in the Works;

## 4 Technical Procedures and Standards

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### 4.1 General

Design, materials, finishes and workmanship will comply with the following:

- UK Building Regulations;
- Current British and European Standards;
- Any additional requirements as otherwise stated by this specification;

### 4.2 BREEAM

The 60 Curzon Street project is required to achieve BREEAM New Construction 2011 rating of 'Very Good' for the restaurant.

### 4.3 Code for Sustainable Homes

The building will achieve a Code for Sustainable Homes (CFSH) Level 4 for the residential areas.

### 4.4 Calculations, Computer Analyses and Test Reports

#### 4.4.1 Calculations generally

Calculations prepared the Sub-contractor to justify his designs for the Works shall be readable, comprehensively tabulated and cross referenced. A calculation issue register shall be maintained, which lists the full set of calculations provided for the project, dates of submission and revision numbers. Calculations shall be done in accordance with appropriate recognised standards, codes of practice and procedures, using diagrams and extracts from drawings to explain what is being analysed. Basic data used and all assumptions made must be clearly stated. Evidence that the engineer responsible for the calculations has appropriate qualifications and is a chartered member of an appropriate recognised professional institution shall be provided. Evidence of internal checking of the calculations by the sub-contractor shall also be provided.

#### 4.4.2 Structural calculations

Structural calculations shall include the following:-

- All load cases, load factors and combinations of load cases including dead loads, imposed loads and barrier loads where appropriate.
- Where appropriate, assessment of overall stability and resistance to progressive collapse.
- Details of assumptions for stress limits for factored or un-factored loads as appropriate for different materials including references to codes giving such values.

- Description or diagram of load paths to explain the structural system, which sets out the detailed structural calculations to follow.
- Where finite analysis models are used, identification of restraint types, property types, axis systems and applied loads. Checks are also required to demonstrate that the global reactions are in accordance with the loading assumptions.
- Tables of reactions.
- Reference to the number and revision of the drawings that were used to prepare the calculations.

Calculations for brackets incorporating movement allowance shall anticipate the most onerous position for the application of loads.

### 4.4.3 Assessing thermal performance

#### 4.4.3.1 Thermal transmittance (U-value)

The heat transferred through windows, curtain walling and other thermally complex elements shall be assessed by 2 dimensional (or 3 dimensional for locations where structure penetrates the cladding) numerical analysis. The software shall be shown to be validated against the examples given in Annex D of EN ISO 10077-2. The results of the assessment shall be summarised as thermal transmittance (U-value) for the centre of the panels, edge effects and frames of each relevant window, curtain walling and metal cladding type. Overall U values for each relevant window, curtain walling and metal cladding type shall be calculated as a weighted average of constituent areas.

The calculation method shall take into account the internal and external air temperatures and surface heat transfer coefficients. These coefficients and the parameters, which affect them, shall be quoted. The method shall take into account the effects of any air spaces and the thermal interaction between elements of the façade.

The calculation should be carried out in accordance with the requirements of BS EN ISO 12631. Guidance can be found in the Centre for Window and Cladding Technology's Guide to Good Practice for assessing glazing frame U-values, Guide to Good Practice for assessing heat transfer and condensation risk for a curtain wall and, Thermal assessment of window assemblies, curtain walling and non-traditional building envelopes.

When applicable, linear and/or point thermal transmittances shall be calculated for thermal bridges through the façade, in accordance with BS EN ISO 10211-1.

The Sub-contractor shall verify that his calculated U-values are compliant with U-values assumed in the Simplified Building Energy Model (SBEM). He shall do this by submitting his U-value calculations for review.

#### 4.4.3.2 Condensation

The risk of condensation shall be assessed in accordance with the requirements of BS EN ISO 13788 with internal and external boundary conditions as specified, and the Centre for Window and Cladding Technology's Guide to Good Practice for assessing heat transfer and condensation risk for a curtain wall and, Thermal assessment of window assemblies, curtain walling and non-traditional building envelopes.

#### 4.4.4 Solar and light performance

Assessing compliance of proposed glazing with specified performance requirements shall be based on the methods described in BS EN 410. Calculations shall be based on the spectral properties of single glass panes.

#### 4.4.5 Computer analyses

Where computer analyses are used the following information shall be presented:

- A clear statement of the method of analysis.
- Proprietary program used.
- The nature of the program e.g. finite element, linear flow, steady state, standard compliance etc.
- Where used, a clear statement of whether finite analysis is linear or non-linear.
- The input data.
- All assumptions made and used.
- The output data.
- A clear statement of the method of interpretation of the results.
- The interpretation of the results.

#### 4.4.6 Test reports

All test reports shall include details of:

- The nature and objectives of the test, measurement or chemical analysis.
- The credentials of the organisation undertaking the test including their accreditation for such testing, measurement or analysis.
- The dates, weather and witnesses present.
- Details of the samples, the method used and the limits of accuracy.
- A factual account of the results (N.B. not an opinion on the results interpretation).

## 5 Performance

### 5.1 Design life

The cladding has a design life as defined in BS ISO 15686-1 of 60 years.

| Design life of building                                                                  | Inaccessible or structural components | Components where replacement is expensive or difficult <sup>a</sup> | Major replaceable components | Building services |
|------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------|------------------------------|-------------------|
| Unlimited                                                                                | Unlimited                             | 100                                                                 | 40                           | 25                |
| 150                                                                                      | 150                                   | 100                                                                 | 40                           | 25                |
| 100                                                                                      | 100                                   | 100                                                                 | 40                           | 25                |
| 60                                                                                       | 60                                    | 60                                                                  | 40                           | 25                |
| 25                                                                                       | 25                                    | 25                                                                  | 25                           | 25                |
| 15                                                                                       | 15                                    | 15                                                                  | 15                           | 15                |
| 10                                                                                       | 10                                    | 10                                                                  | 10                           | 10                |
| NOTE 1 Easily replaced components may have design lives of three to six years.           |                                       |                                                                     |                              |                   |
| NOTE 2 An unlimited design life should rarely be used as it significantly reduces design |                                       |                                                                     |                              |                   |
| <sup>a</sup> Including below-ground drainage.                                            |                                       |                                                                     |                              |                   |

The predicted service life of components will be the service life predicted from recorded performance over time, as defined in clause 3.1.5 of BS ISO 15686-1

All primary components, as defined in section 7.2.4 of the CWCT Standard for Systemised Buildings, will have a predicted service life of not less than the design life of the cladding, without the need for any cleaning or maintenance.

Secondary components as defined in section 7.2.5 of the CWCT Standard for Systemised Buildings which, under the specified conditions, have a predicted service life that is less than the design life of the cladding, will be identified and brought to the attention of the Architect.

Secondary components will be capable of replacement without compromising the long-term structural or weatherproof integrity of the cladding system. They will be capable of replacement without progressive dismantling of the cladding system.

### 5.2 Warranties

Warranties will be provided for the works and its components in accordance with the Contract. Specific Warranties for façade components shall be provided in accordance with the table below.

| Nr | Item                              | Warranty period (years) |
|----|-----------------------------------|-------------------------|
| 1  | Insulating Glass Units            | 10 Years                |
| 2  | Unitised framing aluminium facade | 25 Years                |
| 3  | Gaskets (Type: Neoprene, EPDM)    | 10 Years                |
| 4  | Polyester Powder Coat surface     | 10 Years                |
| 5  | Structural Sealant                | 10 Years                |
| 6  | Anodised Surface                  | 25 Years                |
| 7  | Stainless Steel                   | 25 Years                |
| 8  | Single Glass                      | 10 Years                |
| 9  | Weather Sealants                  | 10 Years                |
| 10 | Insulation                        | 10 Years                |
| 11 | Ironmongery                       | 5 Years                 |
| 12 | Membranes and Vapour Barriers     | 10 Years                |
| 13 | Motors and other electrical items | 1 Year                  |
| 14 | Render system                     | 10 years                |

## 5.3 Applied loads

### 5.3.1 Gravity

Façades will support their self-weight and transmit this weight safely back to the supporting structure without overloading or permanently displacing any of its components.

## 5.3.2 Wind

### 5.3.2.1 General

Facades will support wind loads and transmit them safely back to the supporting structure without overloading or permanently deforming any of its components.

#### 5.3.2.2 Wind resistance-safety

When tested at both positive and negative applications of the design wind loads, multiplied by a factor of 1.5, there will be no permanent damage to framing members, panels or anchors. Framing members will not be buckled. Panels, glazing beads, structural silicone assemblies and decorative capping pieces will remain securely held and gaskets will not be displaced. Opening vents will remain closed and will not be disengaged from their frames.

Residual deformation to framing members will not exceed 1/500 of the span measured between points of attachment to the structure one hour after loading has been removed.

Displacement of framing members at supports will not exceed 3.0mm taken as the difference between the position under peak positive load and peak negative load.

## 5.3.3 Snow/ ice

The façade will support loads from snow and ice and transmit these safely back to the supporting structure in accordance with BS 6399 Part 3.

## 5.3.4 Cleaning

The Works will sustain safely and without damage, a static load of 500 N applied horizontally through a square of 100 mm side on any part of the framework, surface metalwork, stone or glass.

## 5.3.5 Handrails and barriers

Where the Works support handrails or provide barriers it will sustain safely and without damage, the following horizontal imposed loads given in Table 4 of BS 6399-1. They will also be designed for the vertical loads given in clause 10 of BS 6399-1.

|                                                                      |                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cladding Types 1A & 1B<br><i>Domestic and residential activities</i> | Line load of 0.74kN/m acting at a height of 1.1m above finished floor level<br>Uniformly distributed load of 1.0kN/m <sup>2</sup><br>Point load of 0.5kN acting on any part up to and including 1.1m above finished floor level |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                 |                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Terraces where people may congregate less than 3.0m wide and susceptible to overcrowding</i> | Line load of 1.5kN/m acting at a height of 1.1m above finished floor level<br>Uniformly distributed load of 1.5kN/m <sup>2</sup><br>Point load of 1.5kN acting on any part up to and including 1.1m above finished floor level |
| <i>Terraces greater than 3m wide susceptible to overcrowding</i>                                | Line load of 3.0kN/m acting at a height of 1.1m above finished floor level<br>Uniformly distributed load of 1.5kN/m <sup>2</sup><br>Point load of 1.5kN acting on any part up to and including 1.1m above finished floor level |

Glass balustrades, and glazing providing containment, will resist a soft body impact energy equivalent to a drop height of 1200mm when impacted with the impactor described in BS EN 12600. When all the leaves of the glass panes are broken, the glass elements will remain attached around their perimeter and no fragment of glass with a dimension greater than 15mm will fall out of element, for a period of 24 hours. In the event that testing is required to verify design, the actual detail will be tested vertically using a test method similar to that described in BS EN 12600.

## 5.3.6 Impact

### 5.3.6.1 General

The Works, excluding window framing profiles, glazed terracotta and stone shall satisfy the following requirements when tested in accordance with CWCT Technical Note 76:

| Test           | Exposure Categories<br>(Table 3 of CWCT<br>Technical Note TN 75) | Impact Energy<br>(Joules) | Performance (Tables 1<br>& 2 of CWCT Technical<br>Note TN 75) |
|----------------|------------------------------------------------------------------|---------------------------|---------------------------------------------------------------|
| Serviceability | B, E & F                                                         | 120                       | Class 1                                                       |
| Safety         | B                                                                | 500                       | Negligible risk                                               |
|                | E & F                                                            | 350                       | Negligible risk                                               |

Punched windows will achieve the following impact performance levels when tested in accordance with BS EN 13049:

External impact: BS 6375-2, Table 2: Test Class 4;

Internal impact: BS 6375-2, Table 2: Test Class 4;

### 5.3.6.2 Doors – soft and hard body impact

External and internal surfaces of doors will be the following classes, in accordance with BS EN 1192 when tested in accordance with BS EN 949 and BS EN 950.

| Cladding Type | Classification (BS EN 1192) – Table A1 |
|---------------|----------------------------------------|
| Types 1A & 1B | 1-2                                    |
| Types 3 & 4   | 2-3                                    |

After hard body impact testing, the mean value of the diameters of the indentations left by any hard body impact will not exceed 20mm and the mean value of the depth of the indentations will not exceed 1.0mm and the maximum value will not exceed 1.5mm.

### 5.3.6.3 Stonework – soft body impact

Stone cladding shall satisfy the requirements in clause 5.3.6.1.

The following locations shall be impacted:

- the centre of the stone slab module with the largest area;
- the centre of the stone slab module with the largest aspect ratio (the ratio of the long side to the short side);
- the centre of the longest horizontal edge;
- the centre of the longest vertical edge;
- the four-way junction.

### 5.3.6.4 Internal glazed areas – soft body impact

All internal glazing including glazing to doors, in critical areas as defined in Section 1.1 of Building Regulations Approved Document N1, will be Class 1 safety glass in accordance with BS EN 12600.

### 5.3.6.5 Non-glazed internal areas of doors in the works

The non-glazed internal areas of any doors in the Works will have the same performance requirements as the external non-glazed parts of doors in the Works.

### 5.3.6.6 Stone soffits

Stone soffits shall resist localised hard body impacts up to 40 Nm impact energy due to maintenance access equipment. When impacted the structural safety of the soffit will not be put at risk, and no part of it will fall or to cause injury to people below the soffit. Any panels which break shall not fall, and if any damage does occur no part of the panel will become detached for sixty minutes after the impact. The impactor shall not pass through the stone.

### 5.3.6.7 Extruded Glazed Terracotta

A soft body impact test will be carried out on a glazed terracotta test samples in accordance with CWCT Technical Notes TN75 and TN76. The impact energy will be as described in table 4 of TN75. The glazed terracotta is in impact category E according to table 3 of TN 75 and as such requires safety impact test but does not require serviceability tests.

These tests are expected to be undertaken by the appointed hand set terracotta Façade Contractor by an approved UKAS accredited body.

For each mock-up the following points will be impacted:

- a) the centre of the glazed terracotta panel with the largest area;
- b) the centre of the glazed terracotta panel with the largest aspect ratio (the ratio of the long side to the short side)
- c) the centre of the longest horizontal edge;
- d) the centre of the longest vertical edge;

#### **5.3.6.8 Impact loads for Rooflight and Canopies**

Overhead rooflight glazing and its assembly and supporting framework will be Class 1 in accordance with CWCT Technical Note 66 when tested at 40 °C.

#### **5.3.7 Point load on glazed areas of canopies**

The canopy will be designed to withstand a 1.0kN concentrated vertical load applied anywhere on its top surface through a square contact area of 100 mm without overloading, dislodgement, permanent deformation or damage.

#### **5.3.8 Thermal loads**

Glass will not break as a result of shading stresses including that from adjacent buildings, from internal glare and black-out blinds / curtains and from furniture.

#### **5.3.9 Temporary conditions**

The façade will sustain any temporary loading that may arise from lifting, storing, transporting, hoisting and installing them, which will induce stresses greater than those which will be induced when the façade is completed.

#### **5.3.10 Access and maintenance**

Any gutters, copings or flat and near flat roof surfaces in the Works that are used for access will sustain the loads given in BS 6399 -3, clause 4.2 safely and without damage.

Any gutters, copings or flat and near flat roof surfaces with no access except for maintenance and cleaning will sustain the loads given in BS 6399-3, clause 4.3.1 safely and without damage.

#### **5.3.11 Occupants**

Internal ledges and framing members will sustain safely and without damage, a vertical load of 1kN or a uniformly distributed load of 0.6kN/m<sup>2</sup>, whichever is the most onerous.

#### 5.3.11.1 Internal loads on insulating glass units

Insulating glass units will not break under the combined effects of temperature, altitude, atmospheric pressure and other specified loads.

The outer panes of insulating glass units will be stiffer than the inner panes in order to minimise distortions in the external reflections from the glass as a result of atmospheric loads.

#### 5.3.12 Combination of loads

Cladding Types will be designed for the loading combinations required by current Codes of Practice/ Standards.

### 5.4 Movement/ deflections

#### 5.4.1 General

Façades will accommodate the following without any reduction in performance:

- Deflections under repeated cycles of the design wind loads;
- Deflections due to design dead loads and specified live loads;
- Changes in dimension and shape arising from specified building movements;
- Changes in dimension and shape of the façade arising from thermal and moisture movement and shrinkage;

#### 5.4.2 Deflection limits

##### 5.4.2.1 Windows

Windows will be Class A in accordance with BS EN 12210.

At wind pressures P1 and P2:

- There will be no visible failures when viewed by normal or corrected vision at a distance of 1m in natural light;
- The test specimen will remain functional and the air permeability tests after tests P1 and P2 will not exceed the upper limits of the specified air permeability class by more than 20%;
- Deflections will be recovered fully when the loads are removed;

##### 5.4.2.2 Stone

Framing members generally – at both positive and negative applications of the peak out of plane load, the maximum horizontal deflection will not exceed 1/200 of the span of the member measured between points of attachment to the building, or 20 mm, whichever is the lesser.

Framing members to which natural stone is to be fixed – at both positive and negative applications of the peak out of plane load, the maximum horizontal deflection will not exceed  $1/360$  of the length measured along the stone edge, or 3mm, whichever is the lesser.

The downward vertical deflection of framing members that support the weight of stone will not exceed  $1/360$  of the length measured along the stone edge.

#### 5.4.2.3 Extruded glazed terracotta

Framing members generally – at both positive and negative applications of the peak out of plane load, the maximum horizontal deflection will not exceed  $1/200$  of the span of the member measured between points of attachment to the building, or 20 mm, whichever is the lesser.

Framing members to which natural stone is to be fixed – at both positive and negative applications of the peak out of plane load, the maximum horizontal deflection will not exceed  $1/360$  of the length measured along the stone edge, or 3mm, whichever is the lesser.

The downward vertical deflection of framing members that support the weight of stone will not exceed  $1/360$  of the length measured along the stone edge.

#### 5.4.2.4 Metal Panels

Under the action of the most onerous combination of loads, the maximum limit for the centre panel deflection of metal panels will be  $\text{span}/90$ , where the span is the length of the longest diagonal dimension between opposite corners.

#### 5.4.2.5 Barriers

Deflection limits will be in accordance with the requirements of BS 6180.

### 5.5 Weathertightness

#### 5.5.1 Air permeability

##### 5.5.1.1 The Building

The overall air permeability of each Residential Block will not exceed  $3\text{m}^3/\text{hr}/\text{m}^2$  at a pressure of +50Pa.

##### 5.5.1.2 Rooflight

At the peak test pressure the permissible air infiltration rate will not exceed  $1.5\text{m}^3/\text{hr}/\text{m}^2$ . The peak test pressure will be as required to achieve Class A4 in accordance with BS EN 12152.

##### 5.5.1.3 Fully rebated windows and doors

At test pressure the permissible air infiltration rate through the opening joints will not exceed  $2.5\text{m}^3/\text{hr}/\text{m}$ . The peak test pressure will be as required to achieve Class 4 in accordance with BS EN 12207.

Non-fully rebated doors will have an air permeability performance commensurate with the architectural design of the doors.

#### 5.5.1.4 Doors with “flush” access thresholds

At peak test pressure the permissible air infiltration rate through the opening joints will not exceed 7.4m<sup>3</sup>/hr/m. The peak test pressure will be as required to achieve Class 4 in accordance with BS EN 12207.

#### 5.5.1.5 Interfaces

Interfaces between stone facade, windows, doors and other works will not be more than 1.5m<sup>3</sup>/hr per metre length of the perimeter of the interface at the maximum pressure for the specified class.

### 5.5.2 Water penetration resistance

#### 5.5.2.1 General

All joints within the Works will maintain their watertightness under the loads and movements specified.

The drainage of water along edge seals of double glazed units will not be permitted unless written confirmation is provided by the unit manufacturer stating that the design life, required service life and warranty, of the unit is unaffected.

#### 5.5.2.2 The Works

The Works including all joints between it and other elements of work, will prevent leakage of water into the interior of the building from the outer face of the assembly, under the action of wind pressure kinetic energy, gravity, surface tension, or capillary action. It will also prevent water entering into those parts of the cladding that would be adversely affected by the presence of water.

#### 5.5.2.3 Skylights

At all pressures up to and including the peak test pressure, there will be no water leakage onto the internal face of the skylight. The peak test pressure will be as required to achieve Class R7 in accordance with BS EN 12154.

#### 5.5.2.4 Fully rebated windows

At all pressures up to and including the peak test pressure, there will be no water leakage onto the internal face of the windows. The peak test pressure will be as required to achieve Class 9A in accordance with BS EN 12208.

#### 5.5.2.5 Doors with “flush” access thresholds

At all pressures up to and including the peak test pressure, there will be no water leakage onto the internal face of the doors. The peak test pressure will be as required to achieve Class 8A in accordance with BS EN 12208.

### 5.5.2.6 Interfaces

Interfaces between stone, punched windows, doors and other works will not leak at the maximum pressure for the specified curtain wall, window and door classes.

## 5.6 Thermal performance

The Works will achieve compliance with The Building Regulations Approved Document L2A 2010.

### 5.6.1 Thermal transmittance

The Works will provide average U-values equal or better than the values given below:

| Cladding Types                                                                                                                                           | Overall U-Value (W/m²K)                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Type 1 & 2 – Windows and doors                                                                                                                           | Refer to appendix A                                                              |
| Type 2 – Rooftop Pavilion opaque facade                                                                                                                  | 0.26                                                                             |
| Type 2 – Rooftop Pavilion clerestory glazing                                                                                                             | 1.5                                                                              |
| Type 3 – Glazing to core area.                                                                                                                           | 1.5                                                                              |
| Type 4 – Glazing to garden unit entrances.                                                                                                               | 2.0                                                                              |
| Type 5 – Glazing to main entrance (inner skin)                                                                                                           | 0.92                                                                             |
| Type 6 – Glazed rooflight (Wintergarden) (*)                                                                                                             | 2.0                                                                              |
| Type 9 – Metal soffit panels                                                                                                                             | 0.15                                                                             |
| Type 10 & 11 – Hand set Stone & Terracotta facades (**)                                                                                                  | 0.18 (Section 3A)<br>0.21 (Section 3B)<br>0.24 (Section 3C)<br>0.11 (Section 3D) |
| (*) Please note that winter garden is outside the thermal line.<br>(**) For locations refer to Architect's drawing series 1137-A-085-01 to 07 inclusive. |                                                                                  |

### 5.6.2 Condensation

#### 5.6.2.1 General

Condensation will not form on, or behind the building side of the vapour control layer, or on surfaces or in areas which compromise the performance of the Works under the conditions specified by the M&E engineer.

| External Conditions |                      |                      |
|---------------------|----------------------|----------------------|
|                     | Dry bulb temperature | Wet bulb temperature |
| Summer              | 32 °C                | 23 °C                |
| Winter              | -3 °C                | -3 °C                |

| Internal Conditions for Apartments       |                               |            |
|------------------------------------------|-------------------------------|------------|
| Room                                     | Room Design Temperature       |            |
|                                          | Winter                        | Summer     |
| Kitchen                                  | 22 ± 2°C                      | 23 ± 2°C   |
| Living Areas                             | 22 ± 2°C                      | 23 ± 2°C   |
| Bedrooms                                 | 20 ± 2°C                      | 23 ± 2°C   |
| Bathrooms/En-suites                      | 23 ± 2°C                      | No cooling |
| WC                                       | 20°C minimum (winter heating) | No cooling |
| Halls                                    | 20 ± 2°C                      | No cooling |
| Lobbies                                  | 20 ± 2°C                      | No cooling |
| Dressing rooms (7m <sup>2</sup> & above) | 22 ± 2°C                      | 23 ± 2°C   |
| Dressing rooms (under 7m <sup>2</sup> )  | 22 ± 2°C                      | No cooling |
| Study                                    | 22 ± 2°C                      | 23 ± 2°C   |
| Utility rooms                            | 18°C minimum (winter heating) | No cooling |
| Stores/cupboards                         | No heating                    | No cooling |

| Internal Conditions for Landlord Areas |                               |            |                              |
|----------------------------------------|-------------------------------|------------|------------------------------|
| Room                                   | Room Design Temperature       |            |                              |
|                                        | Winter                        | Summer     |                              |
| Stairs                                 | 18°C minimum (winter heating) | No cooling | TBC with mechanical engineer |
| Lift lobbies                           | 18 ± 2°C                      | 24 ± 2°C   | TBC with mechanical engineer |
| Car parking                            | No heating                    | No cooling | TBC with mechanical engineer |
| Common W/Cs                            | 20°C minimum (winter heating) | No cooling | TBC with mechanical engineer |
| Reception Areas                        | 20 ± 2°C                      | 23 ± 2°C   | TBC with mechanical engineer |
| Plant Areas                            | Frost control                 | No cooling | TBC with mechanical engineer |

| Internal Conditions for Health Club |                                      |            |
|-------------------------------------|--------------------------------------|------------|
| Room                                | Room Design Temperature              |            |
|                                     | Winter                               | Summer     |
| Treatment room                      | 24 ± 2°C                             | 23 ± 2°C   |
| Meditation area                     | 24 ± 2°C                             | 23 ± 2°C   |
| Changing area                       | 24 ± 2°C                             | No cooling |
| W/Cs                                | 24 ± 2°C                             | No cooling |
| Swimming pool                       | As required to suit Pool Environment |            |

| Internal Conditions for Restaurant |                               |            |
|------------------------------------|-------------------------------|------------|
| Room                               | Room Design Temperature       |            |
|                                    | Winter                        | Summer     |
| Dining Area                        | 21 ± 2°C                      | 23 ± 2°C   |
| Commercial Kitchen                 | 19 ± 2°C                      | 23 ± 2°C   |
| Common W/Cs                        | 20°C minimum (winter heating) | No cooling |

RH (Relative Humidity) will not be directly controlled although in summer beneficial dehumidification will take place when the cooling units are operating.

There will be no winter humidification.

#### 5.6.2.2 Additional conditions taking into account night sky radiation for non-vertical glazing

External conditions will be assessed in accordance with the requirements of BS EN 15927.

### 5.6.3 Solar and light performance

The glass areas will provide solar control values equal or better than the following:

| Cladding Type                   | Visible Light Transmittance | External Reflectance | G-value |
|---------------------------------|-----------------------------|----------------------|---------|
| All windows/doors with TGU      | 0.76                        | 0.15                 | ≤0.54   |
| All windows/doors with DGU      | 0.82                        | 0.12                 | ≤0.61   |
| Glazed rooflight (Wintergarden) | TBC                         | TBC                  | TBC     |

## 5.6.4 Evaluation of colour of coated glass

When measured by an Independent Testing Laboratory using a properly calibrated colorimeter following the CIE  $L^*a^*b^*$  colour space system, the  $\Delta L^*$ ,  $\Delta a^*$  and  $\Delta b^*$  values are expected to be as follows:

Colour in reflection:

|              |     |
|--------------|-----|
| $\Delta L^*$ | 3.0 |
| $\Delta a^*$ | 2.0 |
| $\Delta b^*$ | 3.0 |

Colour in transmission:

|              |     |
|--------------|-----|
| $\Delta L^*$ | 4.5 |
| $\Delta a^*$ | 4.5 |
| $\Delta b^*$ | 4.5 |

All measurements will lie within one quadrant of the CIE  $L^*a^*b^*$  colour space.

Laboratory and site measurements will be carried out for colour in reflection and transmission, for both faces of the glass units, in accordance with the requirements of ASTM C 1649 and ASTM C 1650.

Laboratory measurements for every combination of glass proposed will be carried out and mean values established and presented to the Architect for review.

Before carrying out any site measurements, the appointed Façade Contractor is expected to agree locations, areas and glazing types to be measured with the Architect

The guidance given in CIE Technical Report 130/1998; Practical methods for the measurement of reflectance and transmittance will be followed.

## 5.7 Acoustic performance

### 5.7.1 Airborne sound insulation

The minimum sound insulation for the façade elements will be in accordance to the Acoustic consultants Stage 4 report. Please refer to this document.

- Solid facades: for all areas an overall noise attenuation of  $R_w$  50dB will be required.
- Glazed facades – Garden: for these areas an overall noise attenuation of  $R_w$  34dB will be required.
- Glazed Facades – Street sides: for the facades overlooking Bolton St and Curzon St an overall noise attenuation of  $R_w$  46dB.

- Glazed Facades – Restaurant: for the restaurant overlooking Bolton St and Curzon St an overall noise attenuation of  $R_w$  37dB will be required.

Note: the performance requirements above are weighted values for reference only. Please refer to the acoustic consultant's Stage 4 report for the requirements at specific frequencies.

### 5.7.2 Demonstration of acoustic performance

Building elements will be tested in accordance with BS EN ISO 10140-2:2010. Independent testing at a UKAS accredited, or international equivalent, laboratory will be required.

Test samples will be representative of the relevant façade types including glass, framing, seals, opening lights, etc.

### 5.7.3 Flanking sound

See Acoustic Consultants report.

### 5.7.4 Seals

The standard of sealing of the works will ensure no reduction in sound insulation performance for the service life.

Gaskets and seals used to achieve the required weather and airflow resistance will be selected to accommodate fully the range of dimensional tolerances associated with fabrication, installation and movement in situ of the works sufficient to maintain the acoustic performance required. Materials for gaskets and seals will be capable of maintaining their elastic qualities, dimensions and resistance to physical or chemical attack.

## 5.8 Noise

The façade will not creak, rattle or whistle due to the wind and/ or specified thermal and structural movements.

### 5.8.1 Anti-drumming treatment

The pavilion roof tiles shall incorporate an anti-drumming treatment to mitigate rainfall noise. The anti-drumming treatment shall be equivalent to “Tecsound 50” or “Raincheck BB93/HTM 08 01”, subject to approval.

## 5.9 Fire performance

### 5.9.1 General

The Works, including junctions with other works, and fire breaks that may be incorporated to meet the performance requirements, will not be composed of materials that readily support combustion and add significantly to the fire load.

Insulation material used in external walls will comply with the requirements of Building Regulations Approved Document B.

### 5.9.2 Surface spread of flames

Materials used internally and externally will have Class O fire classification as defined in Approved Document B to the Building Regulations.

### 5.9.3 Compartmentation

Where the facade abuts compartment floors it will be restrained at floor level to limit the movement of the facade away from the compartment floor when exposed to fire.

### 5.9.4 Fire and smoke stopping

The Works will comply with the relevant fire resistance and fire stopping recommendations of Approved Document B to the Building Regulations, and BS 476: Parts 20+22.

### 5.9.5 Cavity Barriers

Provide cavity barriers of 30 minutes fire resistance (E30 and I15) at the junction between any external cavity wall and every compartment floor and compartment wall, in accordance to Approved Document B to the Building Regulations, and BS 476: Parts 20+22.

### 5.9.6 Smoke venting

The method of opening the smoke vents will be acceptable to the fire service.

Smoke vents will comply with Section 4.4 of the CWCT Standard for systemised building envelopes and clause 5.14 of this specification

Window hardware and stays will be provided in accordance with the Architect's window schedule.

## 5.10 Doors

### 5.10.1 Security Classification

The security performance of all doors shall be in accordance with the principles of ‘Secured by Design’. All doors directly accessible to the public shall be rated as SR2 in accordance with LPS 1175.

### 5.10.2 Strength, mechanical durability

In addition to requirements noted elsewhere in this specification, in particular Section 5.3.5 regarding impact performance, doors will comply with the requirements of Section 6 and Table A2 of Annex A of BS 6375-2:2009.

| Cladding Type | Classification (BS EN 1192) – Table A1 |
|---------------|----------------------------------------|
| Types 1 & 2   | Class 1                                |
| Types 4 & 7   | Class 2                                |

### 5.10.3 Operating forces

The requirements of Table A2 of Annex A of BS 6375-2 will be followed.

### 5.10.4 Resistance to repeated opening and closing

When tested in accordance with BS EN 1191, doors will achieve the following classes given in BS EN 12400:

| Cladding Type | Classification (BS EN 1192) – Table A1 |
|---------------|----------------------------------------|
| Type 1 & 2    | Class 3                                |
| Type 4 & 7    | Class 5                                |

### 5.10.5 Controlled door devices

Controlled door closing devices will comply with BS EN 1154. Electrically powered hold-open devices for swing doors will comply with BS EN 1155.

## 5.11 Opening windows

In addition to requirements noted elsewhere in this specification, in particular Section 6.3.6 regarding impact performance, opening windows will comply with the requirements of Building Regulations Approved Document N and Section 5 and Table A1 of Annex A of BS 6375-2:2009, with the following exception:

The test load to be used to test the load bearing capacity of safety devices in accordance with BS EN 14609, will be calculated based on 50% of the design wind load acting on the full area of the openable part of the window.

Requirements and test methods for window hardware will be in accordance with BS EN 13126-1 and the appropriate part of BS EN 13126 parts 2 to 17.

The recommendations in BS 8213-1:2004 will be followed in respect of restricting catches and reach distances for operating handles and other devices to ensure safe operation of opening windows by users and cleaners.

## 5.12 Lightning protection

### 5.12.1 General

Cladding types will meet the requirements of BS EN 62305, BS 7671, and BS 7430. No external tapes or visible connections will be accepted.

### 5.12.2 Equipotentiality

Metal component parts in facade will be mechanically connected together with the building structure to provide an equipotential bond to the earth circuit of the building. The electrical resistance of the bond of the curtain wall will not exceed  $10\Omega$  when tested in accordance with annex A of BS EN 13830.

## 5.13 Vibration

Primary brackets and fixings will not loosen due the effects of vibration or the cyclic effects of loads, deflections and thermal movements.

## 5.14 Corrosion

### 5.14.1 General

Materials will be selected and installed, so that they will remain compatible with the other materials around it within its range of influence for the service life of the façade.

Adequate measures will be taken to prevent bimetallic corrosion between dissimilar metals. Reference will be made to PD 6484 'Commentary on corrosion at bimetallic contacts and its alleviation'.

Where different metallic materials are used, separators will be fitted to eliminate the risk of corrosion.

All aluminium or galvanised steel components in direct contact with, but not totally encased in, cementitious surfaces will be isolated with thin dense PVC or EPDM isolation packs. The use of bituminous paint will not be accepted.

## 5.15 Infestation

Materials used in the facades will not be attacked or infested by micro-organisms, fungi or other known vermin.

## 5.16 Security

The security performance of windows will be designed in accordance with the principles of 'Secured by Design'.

## 5.17 Brackets

All brackets outside the vapour control line will be stainless steel or aluminium.

## 5.18 Fixings

Fixings will comply with the requirements of the Building Regulations Approved Document A, Section 2. They will be designed so that there will be no risk of loosening due to specified loads, deflections and thermal movements. All fixings outside or through the vapour control line will be stainless steel.

## 5.19 Accent Lighting

Provide provision for cabling and support of accent lighting through the façade build up as required, refer to External Lighting Consultant Stage D report.

## 6 Verification

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### 6.1 Visual appearance

#### 6.1.1 Samples

Labelled samples of products, components, materials and finishes shall be provided for approval by the Architect. Unless stated otherwise the samples shall be full size or a minimum size of 500mm x 500mm for all flat sheet materials, and minimum 1000mm length of all linear elements such as gaskets, tubes, rails and extrusions. As a minimum this shall include:

- Thermally broken anodised aluminium window and door profiles.
- Fabricated anodised aluminium cills, soffit and closure panels.
- Anodised aluminium louvre blades.
- Bespoke anodised aluminium extrusions.
- Fabricated anodised aluminium trims/closures (party wall interfaces).
- Bronze and patinated stainless steel balustrade elements.
- Insulation and membranes.
- Mortar and sealants.
- Gaskets.
- Visible fixings.
- Ironmongery, window and door hardware.
- Glass units.
- Stone cladding, including rusticated features.
- Terracotta, including bespoke glazed terracotta tiles.
- Onyx cladding.

The contractor is additionally expected to supply repair samples for all finishes.

##### 6.1.1.1 Glass

Glass samples provided for visual verification shall be a minimum of 1000mm x 1000mm and produced on the same production lines as those proposed for the final production runs.

Full-size samples will be provided to assess transmission and reflection under a variety of natural and artificial lighting conditions.

### **6.1.1.2 Bespoke glazed terracotta tiles**

A series of three sample production runs/batches shall be provided for the bespoke glazed terracotta tiles. Each sample production run/batch shall include 6no. flat (500X500mm) tiles and 1no. three dimensional “coin”. The Architect shall review and each sample production run/batch.

### **6.1.1.3 Stone**

Stone range samples that include all the features which, in the opinion of the stone supplier, are likely to occur Refer to section 7.10.3.

## **6.2 Off-site visual mock-ups**

### **6.2.1.1 General requirements**

The appointed Façade Contractor is expected to construct, in his yard or at an agreed location, visual mock-ups as described on the Architect’s drawings. Please refer to drawings 499-02 and 499-03.

These mock-ups will, unless otherwise agreed by the Architect, be constructed in the orientation that they will be in the finished Works on temporary supporting structures that will allow inspection of internal and external surfaces. They will, unless otherwise agreed by the Architect, be placed outside so that the effects of sunlight and shade can be assessed.

Components are only required to show the internal and external appearance, arrangement and dimensions. Bespoke extrusion profiles will not be required and can be simulated with painted timber assemblies or similar. Standard extrusions profiles shall be used where applicable. The glass proposed in the Works and current colours of elements as proposed by the Architect will be required. The intention is to agree the visual appearance of the as-installed glass, to show the relationship between the glazing, metalwork framing and their joints, etc., and will be produced to good time to allow any final minor adjustments to be made to the design prior to fabrication.

## **6.3 Weather performance testing**

### **6.3.1 Off-site weather performance tests**

#### **6.3.1.1 General**

2 No. off-site weather performance tests will be carried in accordance with The Centre for Window and Cladding Technology (CWCT), Standard Test Methods for Building Envelopes. Performance mock-ups as described on the Architect’s drawings. Please refer to drawings 499-04 and 499-05.

If the test fails at any stage, only 2 retests shall be allowed before the rig shall be dismantled and rebuilt

#### **6.3.1.2 Test sequence**

The test sequence will be as follows:

- Air permeability - infiltration;
- Air permeability - exfiltration;
- Water penetration resistance - static;
- Wind resistance - serviceability;
- Air permeability - infiltration;
- Air permeability - exfiltration;
- Water penetration resistance - static;
- Water penetration resistance - dynamic;
- Hose test;
- Wind resistance - safety;
- Impact testing of windows and stonework - soft body;
- Controlled dismantle, inspect and record;

#### **6.3.1.3 Air exfiltration**

At test pressures up to 100Pa, the permissible air exfiltration rate through the curtain wall will not exceed 3m<sup>3</sup>/hr/m<sup>2</sup>.

#### **6.3.1.4 Repeat air permeability tests**

Clause 3.5.2.13 of the CWCT Standard for Systemised Building Envelopes, will not apply. Instead, the performance will be the same or better than that measured during previous tests.

#### **6.3.1.5 Water spray rate for water penetration tests**

The water spray rate will be 3.4l/m<sup>2</sup>min.

#### **6.3.1.6 Dynamic water test – test pressure**

The nominal test pressure will be 600Pa.

#### **6.3.1.7 Dynamic water test – test method**

The dynamic water test method will be in accordance with Section 7 or Section 8 of the CWCT Standard Test Methods for Building Envelopes.

#### **6.3.1.8 Hose testing of the test specimen – areas to be tested**

Joints to be tested will be selected by the Architect on the day of the test.

#### **6.3.1.9 Impact testing of the test specimen**

Impact testing will be carried out in accordance with BS EN 12600 and BS EN 13049 for soft body impact.

## **6.4 Acoustic testing**

### **6.4.1 Airborne sound**

Acoustic testing will be in accordance with BS EN ISO 10140-2 and be carried out in a UKAS accredited, or international equivalent, laboratory.

### **6.4.2 Horizontal and vertical flanking**

Acoustic testing will be in accordance with BS EN ISO 10848-2 and be carried out in a UKAS accredited, or international equivalent, laboratory.

## **6.5 Impact testing**

### **6.5.1 General**

Impact testing will be carried out in accordance with BS EN 12600 and BS EN 950.

### **6.5.2 Robustness testing of rooflights and canopies**

Overhead glass will be tested in accordance with CWCT Technical Note 67.

## **6.6 Materials testing**

### **6.6.1 General**

Tests will be carried out by an Independent Testing Authority who is accredited to carry out the required tests.

### **6.6.2 Stone**

#### **6.6.2.1 Refer to section 7.10 Sealant (non-structural)**

Testing will be carried out to demonstrate that sealant is non-staining and compatible with the natural stones proposed.

### **6.6.3 Extruded glazed terracotta approval testing**

These tests are required to appraise the suitability of the glazed terracotta being supplied for this project, and to ascertain its strength and durability characteristics, informing the design of the glazed terracotta panels. These tests will be carried and the results approved by the Architect prior to commencing manufacture of the terracotta.

Test certificates (from a UKAS accredited laboratory or approved alternative) less than 18 months old that are available for the glazed terracotta type under consideration which demonstrate the material meets the technical requirements in section 4.1 may be submitted

as possible alternatives to undertaking the tests described below, provided they relate to the same ceramic body composition.

#### **6.6.4 Structural silicone glazing**

Testing will be carried out in accordance with either the requirements of the European Organisation for Technical Approvals (ETAG) document 002 or BS EN 13022 Parts 1 & 2.

#### **6.6.5 Mortar**

Mortar to be used in conjunction with the stone will be tested to determine if there is any related staining of the natural stone related to alkali migration from the mortar. A lime mortar staining test will be carried out by an independent laboratory.

#### **6.6.6 Welding**

Welder qualification, procedures and tests will be carried out in accordance with BS EN standards.

### **6.7 Tests to fixings**

#### **6.7.1 General**

Test data, calculations and technical literature will be provided to demonstrate that the proposed fixings are adequate. Where this information is not available an Independent Testing Authority will carry out tests in accordance with BS 5080 Parts 1 and 2 and the Construction Fixing Association Guidance Note: Procedure for site testing construction fixings.

#### **6.7.2 Stone fixings**

##### **6.7.2.1 Dowel fixings**

Tests shall be carried out in accordance with BS EN 13364.

Ten specimens taken from each block. The specimens shall be tested in the orientation representative of their in-service performance.

##### **6.7.2.2 Kerf fixings**

Tests shall be carried out in accordance with ASTM C 1354.

Ten specimens taken from each block. The specimens shall be tested in the orientation representative of their in-service performance.

#### **6.7.2.3 Samples for testing**

The samples used to carry out the tests shall be taken from blocks taken from at least three separate parts of the quarry currently being worked. The same blocks shall be used to provide specimens for the stone property tests described elsewhere in this Contract Document, so that a relationship between the stone properties and the fixing tests can be established. The bedding plane or rift shall be marked on the specimens by the quarry.

#### **6.7.3 Tests for fixing in stone (prior to processing)**

Pull-out tests for stud fixings/ undercut anchors, kerfs and dowel fixings will be carried out.

#### **6.7.4 Tests for fixings into glazed terracotta (prior to processing)**

Pull-out tests for stud fixings/ undercut anchors, kerfs and dowel fixings will be carried out.

#### **6.7.5 Tests for fixing in laminated stone panels (prior to processing)**

Pull-out tests for undercut anchors will be carried out in accordance with ASTM C 1354.

### **6.8 Finishes testing**

#### **6.8.1 Third party testing of finishes**

An Independent Inspection Authority will carry out a minimum of two acceptance inspections for each finish type. Sampling procedures and plans will comply with BS 6001: Part 1 (ISO 2859/2), for general inspection Level 2, AQL = 1%.

### **6.9 Control Benchmarks**

#### **6.9.1 Factory benchmarks**

The first window and door of each window and door type that comes off the fabrication line will be used as the benchmark for quality. Each window and door will be set up outside in the orientation they will be in the finished works. Each Panel will be offered and inspected in the contractors fabrication factory in similar light conditions as the real job and full production shall not commence until these are signed off by the Architect. Each panel will be inspected and 'signed-off' by the Architect.

#### **6.9.2 Site workmanship**

The first area of each cladding type will be offered as an on-site quality control sample to assess the standard of workmanship. A hold point shall be issued at this point requiring

approval to proceed if the contractor proceeds within sign off it is at their own risk. Workmanship below the standard of the accepted sample will be rejected.

### 6.9.3 Site hose tests

Hose testing will be carried out on the following:

- 10% of linear joints of each vertical cladding type;
- 100% of linear joints of the Wintergarden Rooflight;

Testing will be in accordance with the CWCT Standard Test Methods for Building Envelopes Section 9.

### 6.9.4 Flood testing of rooflight/water-feature

Flood testing of the fully installed rooflight is expected to be allowed for. The Independent Testing Authority will supervise the flood testing.

The entire rooflight will be filled to a minimum flood depth of 150mm. The water will be held in the rooflight for a minimum of 24 hours. Measurements of the depth of the water in the rooflight will be taken every hour during the working day. The water level will be maintained at 100mm by topping up from a calibrated closed water tank so that the amount and rate of any water loss can be established.

Areas below and adjacent to the rooflight will be inspected every hour during the working day. The rooflight will be deemed to fail upon any detected leakage during the test and up to one hour after the test sample has been drained.

## 6.10 Whole building air leakage test

Prior to final completion, the air leakage characteristics of the buildings will be tested in accordance with BS EN 13829. The air leakage rates will not exceed the value given in this Specification.

## 7 Materials

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### 7.1 General

Materials will be selected by reference to the guide “Good Practice in the Selection of Materials” available from the British Council of Offices (BCO) and the British Property Federation (BFP).

All materials will comply with current British Standards, European Standards or when noted in this specification to be higher and will be compatible for the design life of the Works.

The appointed Façade Contractor is expected to obtain the total quantity of each product type from the same manufacturer. Any deviation from the above will not be acceptable.

The appointed specialist Façade Contractor will obtain materials from established and reputable manufacturers.

The sourcing of all materials will conform to the BREEAM when required.

#### 7.1.1 Identification of materials

All principal materials delivered to the factory or site are expected to bear the manufacturer's name, brand name and/or any other data that may be required to verify the exact nature of the material and relate it to the requirements of this Contract Document. Where applicable, the material will bear the Works test certificates, British Standards Certification, Trade Mark, CE Mark and/or British Board of Agrément Certificate mark. Test certificates will include, whenever applicable, the location in the Works or the delivery or batch which the sample represents.

#### 7.1.2 Compatibility of materials

The appointed Façade Contractor is expected to select and install each material so that it is, and will remain, compatible with the other materials around it and within its range of influence for the service life of the Works. As described in BS EN ISO 12944-2, the environment is C3.

The appointed Façade Contractor is expected to ensure that adequate measures will be taken to prevent bimetallic corrosion between dissimilar metals. To this end, attention is drawn to publication PD 6484 'Commentary on corrosion at bimetallic contacts and its alleviation'.

Where different metallic materials are used, separators will be fitted to eliminate the risk of corrosion.

### 7.1.3 Unspecified materials

As part of their submissions after appointment, the appointed Façade Contractor is expected to submit a list of all materials they propose to use that are not specifically described in this Contract Document. These materials will conform to the general requirements specified herein.

Unspecified materials will be subject to review by the Employer.

## 7.2 Aluminium

### 7.2.1 Extrusions

All extruded aluminium for framing members shall have the chemical composition and temper appropriate for its function, exposure and applied finish. They shall comply with BS EN 755, BS 1161 and BS EN 515.

Design shall comply with the appropriate parts of BS EN 1999-1-1 to -1-5 inclusive and the UK National Annex to BS EN 1999. The extrusions shall have the webs, walls, and flanges of sufficient thickness to satisfy all structural requirements and eliminate permanent distortion of elements in the finished Works.

### 7.2.2 Sheet, strip and plate

Sheets for hidden flashings will not be less than 1.6 mm thick. Sheets for all components exposed to view or to impact including copings, panels and visible closures will not be less than 4 mm. They shall comply with BS EN 485 and BS EN 573.

## 7.3 Mild steel

## 7.4 Steel Components

### 7.4.1 Mild Steel

The use of mild steel will only be permitted where the component is readily accessible for regular inspection and maintenance.

### 7.4.2 Generally

Steelwork shall be designed in accordance with the appropriate parts of BS EN 1993-1-1 to -1-12 inclusive and the UK National Annex to BS EN 1993.

The use of mild steel shall only be permitted where the component is readily accessible for regular inspection and maintenance.

Where used, it shall be one of the following:

Hot rolled steel complying with the requirements of BS EN 10025.

Hot rolled sections to BS 4: Part 1, BS EN 10210-2, BS EN 10021-1 and BS EN 10297, tubes to BS 6323: Parts 1, 2 & 3 and angles to BS EN 10056-1.

Cold formed pre-galvanised steel sections to BS EN 10143, BS EN 10346.

Steel sheet to BS 1449: Part 1.

Ordinary bolts and nuts to BS EN ISO 4032, BS EN ISO 4033, BS EN 24034.

All plain and tapered washers to BS 4320, Form E.

### **7.4.3 Hot dip galvanising**

Where mild steel sections or fabrications are hot dip galvanised the following additional specification shall apply

Mild steel shall have a carbon equivalent (calculated by the IIW formula) equal to or less than 0.43%

All welding procedures and welding procedure specifications shall be limited to a maximum carbon equivalent of 0.43%.

### **7.4.4 Welding procedures and consumables**

Consumables shall be stored, handled and used in accordance with the manufacturers' recommendations.

Consumables for use in metal arc welding shall comply with BS EN ISO 2560, BS EN ISO 14341, BS EN ISO 17632 or BS EN 756 as appropriate.

## **7.5 Stainless steel**

### **7.5.1 Sheet, strip and plate**

Stainless steel in plate, sheet and strip shall comply with BS EN 10029, BS EN 10048, BS EN ISO 9445 and BS EN 10088 as appropriate.

Grade 1.44xx (formerly 316) will be used for visible components, in all other circumstances grade 1.43xx (formerly 304) will be used.

### **7.5.2 Fixings and fasteners**

Fixings and fasteners shall comply with BS EN ISO 3506-1 and BS EN ISO 3506-2.

Unless noted otherwise, grade A4 shall be used for visible fasteners, in all other circumstances grade A2 shall be used.

The fastener property class shall be selected by the Sub-contractor so as to meet the performance criteria specified herein.

Set screws shall comply with BS EN ISO 3506-3.

Self-tapping screws shall conform to BS EN ISO 3506-4.

### 7.5.3 Sections

Stainless steel sections shall comply with BS EN 10088-3.

### 7.5.4 Welding consumables

Welding consumables shall comply with BS EN ISO 14343 and/or BS EN 1600 as appropriate.

## 7.6 Copper and copper alloys (bronze tiles and brass handrails)

### 7.6.1 General

The appointed Façade Contractor will be responsible for the selection of the copper and copper alloy to suit their proposed manufacturing processes. In addition to the performance criteria for the façade as a whole consideration will be given to the following in selecting the alloy:

- Ease of manufacture
- Strength and ductility
- Durability
- Visual appearance
- Finished appearance; ease of patination and ability to achieve and retain the desired finish colour.

### 7.6.2 Sheet, strip and plate

Copper and copper alloy sheet, strip and plate will comply with BS EN 1172, BS EN 1173, BS EN 1412 and BS EN 1562 as appropriate.

Copper and copper alloy sheet, strip and plate will be of a minimum thickness to meet specified performance criteria without showing signs of either excess deflections or being vulnerable to local impact damage.

### 7.6.3 Profiles and sections

Copper and copper alloy extrusions will comply with BS EN 12163 and BS EN 12167 as appropriate.

#### 7.6.4 Fixings and fasteners

All bronze fixings and fasteners, and securing clips will be compatible with the parent material. The choice of fastener material will be so selected by the appointed Contractor so as to meet the performance criteria specified here in.

#### 7.6.5 Service life

The appointed façade contractor is expected to warrant the bronze finishes will not discolour and are fit for its intended purpose for a minimum of 20 years.

#### 7.6.6 Fabrication requirements

The appointed Façade Contractor is expected to ensure previously agreed uniformity of colour, texture and gloss, on completed elevations and between adjacent panels, within the approved control samples. They will agree a sequence of finishing of the bronze in consistent batches.

#### 7.6.7 Performance criteria

The 'bronze' finishes will satisfy the following:

- All finishes will be stable, fade resistant, and not affected by ultraviolet light. The appointed Façade Contractor will submit all available data and samples on the likely effects of the above for review by the Architect.
- All finishes will be durable and be resilient to all environmental pollution. The appointed Façade Contractor will submit all available data and samples on the likely effects, for review by the Architect
- Minor scratches and blemishes will be repairable using the recommended patination treatment, matching the original finish colour, texture and gloss.
- All finishes will be within the limits of the agreed control samples
- No fixings will be visible in the finished works.
- Run-off from the bronze work will not result in staining of other materials.

#### 7.6.8 Cold sealing

Where required the patination treatment will be subject to cold sealing; e.g. waxing or clear lacquered. The appointed Facade Contractor will provide complete details of any cold sealing techniques to the Architect for review.

#### 7.6.9 Temporary protection

All surfaces that are vulnerable to damage during handling and installation or by subsequent site operations will be fully protected for the duration of the Works.

The appointed Façade Contractor will prevent finished bronze materials from being exposed to any materials which may cause damage to the finish.

### **7.6.10 Remedial works**

Site rectification of damage to the finishes will be subject to the approval of the Architect and will only be permitted subject to submission and approval of a specification, method statement and approved repaired sample. The rectification of damage will not invalidate the warranty on the Bronze Cladding System or its finish.

### **7.6.11 Storage**

All finished bronze are expected to be stored in a manner to prevent damage to the finishes and underlying metalwork.

## **7.7 Glass**

### **7.7.1 General**

Basic soda lime silicate glass shall comply with the provisions of BS EN 572 and BS 952.

The total quantity of each glass material will be obtained from the same material manufacturer. In the case of coated glasses, or other processed glass products, all products will be processed from 'raw' glass material obtained from one manufacturer at the same time.

### **7.7.2 Visual acceptance criteria for basic float glass**

#### **7.7.2.1 General**

All basic float glass shall be clear glass in accordance with BS EN 572-2.

The quality of the glass of the float line for both supplied sizes and final cut sizes shall be assessed for optical and visual faults as described in BS EN 572-8.

Where final cut sizes are supplied, the quality of the cut edge shall be in accordance with Table 19 of BS EN 572-8.

#### **7.7.2.2 Optical faults**

When viewed under the conditions of observation described in Clause 5.2.1 of BS EN 572-2, the angle  $\alpha$  at which there is no disturbing distortion shall not be less than the appropriate critical viewing angle given in Clause 6.3.1 of BS EN 572-8 for supplied sizes and final cut sizes.

### 7.7.2.3 Visual faults

Spot faults shall be no worse than Category C in Table A.1 of BS EN 572-8.

The allowable number of spot faults shall be in accordance with Table 11 of BS EN 572-8 for supplied sizes and Table 16 of BS EN 572-8 for final cut sizes.

### 7.7.2.4 Linear/extended faults

Linear/extended faults shall be in accordance with Clause 6.3.2.4 of BS EN 572-8 for supplied sizes and Clause 6.3.2.5 of BS EN 572-8 for final cut sizes.

## 7.7.3 Sizes

All glass shall be cut to accurate sizes and, if applicable, delivered to site in the required sizes. No on-site cutting or nipping shall be allowed. The glass shall be clearly marked to show its intended final position and orientation.

## 7.7.4 Marking on glass

### 7.7.4.1 Safety glass

Marking on glass providing protection from impact from people shall comply with the requirements of Clauses 7.1 and 7.2 of BS 6262-4.

### 7.7.4.2 Non-safety glass

Non safety glass shall be clearly and indelibly marked with the following information:

- a) The name or trade mark of the manufacturer, merchant or installer;
- b) The identifier of the product standard that the glass conforms to;

## 7.7.5 Thermally toughened glass\*

Thermally toughened glass shall comply with BS EN 12150. Flat glass shall be processed horizontally.

All toughened glass will be heat soak tested in accordance with BS EN 14179. Each pane of heat soak tested toughened glass will be traceable to individual heat soak test batches, in order that, in the event of failure, it will be possible to identify and locate all panels from that heat soaked test batch.

Small shells and / or chips shall be ground out prior to toughening. Any remaining chips and/or shells shall not affect the structural performance of the glass and shall not be visible in the installed Works.

The appearance of anisotropy or 'Leopard Spots', a multi-coloured pattern, is known to be associated with toughened glass under certain polarised lighting conditions. Processing shall be strictly controlled to ensure that this effect is kept to an absolute minimum.

\*The use of fully toughened glass will be avoided unless no alternative is suitable.

### 7.7.6 Heat strengthened glass

Heat strengthened glass will comply with BS EN 1863. Flat glass will be processed horizontally.

Small shells and / or chips will be ground out prior to toughening. Any remaining chips and/or shells will not affect the structural performance of the glass and will not be visible in the installed Works.

The appearance of anisotropy or 'Leopard Spots', a multi-coloured pattern, is known to be associated with heat strengthened glass under certain polarised lighting conditions. Processing will be strictly controlled to ensure that this effect is kept to an absolute minimum. The project shall be reviewed as a whole. If some areas are found to be worse than others they will be rejected.

### 7.7.7 Curved heat treated glass

#### 7.7.7.1 General

Curved glass will be in accordance with ASTM C1464.

#### 7.7.7.2 Bending and heat treating process

The glass will be bent and heat treated on a machine in which the hot glass is kept moving during the quenching stage to avoid hot spots, or zones of slow cooling on its surface.

Any process in which the hot glass is supported on local static contact points during the quenching stage will not be permitted.

#### 7.7.7.3 Residual stress and quality control

The entire surface, including both faces and all edges, of the curved glass panes will be in a state of permanent residual compressive stress. The minimum value of the residual compressive stress at any point on the pane will not be less than that required to comply with BS EN 14179.

Non-destructive measurements of the residual stress, and photographs taken through a polariscope, will be made at sufficient points on each pane to demonstrate that the residual stress is consistent. These will be submitted to the Architect for review in accordance with the Schedule of Submissions. Copies will also be included in the appropriate section of the Operations and Maintenance Manual.

### 7.7.8 Laminated glass and laminated safety glass

Laminated glass and laminated safety glass will comply with BS EN ISO 12543.

Laminated glass and laminated safety glass will meet the durability requirements set out in BS EN ISO 12543-3.

Impact resistance will be as specified elsewhere in this document.

Evaluation of conformity/product standard will be in accordance with BS EN 14449.

Defects in the vision area will be in accordance with clause 4 of BS EN ISO 12543-6.

Defects in the edge area for edges that are framed in the final works will be in accordance with clause 5 of BS EN ISO 12543-6.

Defects in the edge area for edges that are not framed in the final works will be in accordance with clause 8 of BS EN ISO 12543-6.

Vents will not be permitted.

Creases and streaks will not be permitted in the vision area.

The edges of individual panes in a laminated sheet supported on setting blocks will not be stepped.

Clouding and de-lamination is permitted within an edge zone not more than 80% of the thickness of the thickest ply of the laminated glass over the service life of the glass. Any clouding and de-lamination outside this zone will be considered a defect.

## 7.8 Structural sealant

Generally, all structural silicone glazing will be carried out off site in factory controlled conditions in accordance with the requirements of BS EN 13022-2. On-site structural silicone glazing will only be permitted in special circumstances and will be subject to the approval of the Architect.

## 7.9 Sealant (non-structural)

In general, sealant will be selected and applied in accordance with the guidelines contained in BS 6213, BS EN ISO 11600 and CIRIA Guide 178. Sealant will be:

Silicone sealant.

One or two part polysulphide sealant.

As proposed by the appointed Façade Contractor to the appropriate British or European Standard, subject to written confirmation being provided by the manufacturer that his product is being used in appropriate manner in an appropriate situation. It will be compatible with natural stone.

Sealant shall be applied over bond breakers/backer rods, set into the joint to control the depth of sealant and to prevent three-sided adhesion.

Quality control adhesion tests shall be carried out throughout the period of manufacture to monitor the effectiveness of the material and application techniques.

Backing rods will be closed cell polyethylene foam. When closed cell sections are used non-gassed produced will be selected.

The colour of visible sealant used will be as specified on the Architects Drawings and via approved samples.

## **7.9.1 Coated glass**

### **7.9.1.1 Coated glass**

Coated glass shall comply with BS EN 1096.

Processing shall be strictly controlled to ensure even coating, minimum colour variation and no streaking or patching.

Where coatings are applied to thermally toughened glass or heat strengthened glass, these coatings shall not make any anisotropic or “Leopard Spot” effects more obvious.

Performance requirements and colour consistency shall be as specified.

## **7.10 Stone**

### **7.10.1 General**

The following stones are under consideration:

- CT-11.1 and CT-11.2 Portland Basebed Limestone, honed finish
- CT-11.3 White Moleanos Limestone, honed finish
- CT-11.4, CT-11.5 and CT-11.6 Beige Moleanos Limestone, bush-hammered finish
- CT-11.7 and CT-11.8 Angola Black granite, flamed finish

Stone that meets the specified performance and visual criteria will be selected. Specified samples and existing test data will be submitted to the Architect as part of the Tender Submission.

CE marking is required for all stones under consideration.

All European stones will be named /identified in BS EN 12440.

The name, a petrographic description and location of non-European stones must be submitted to the Architect.

Proposals for the stones together with any alternatives that may be considered appropriate will be submitted to the Architect for review.

Stonework will generally be designed in accordance with the requirements of BS 8298 Code of Practice for Design and Installation of Natural Stone Cladding and Lining.

## 7.10.2 Stone properties

### 7.10.2.1 General

Each finished stone element will be sound quality, free from defects (e.g. vents, cracks, holes, seams, fissures, inclusions of deleterious or foreign matter) liable to affect the physical properties, durability or appearance, other than those agreed as acceptable to the Architect from the range samples.

### 7.10.2.2 Mechanical and physical properties and certificates prior to procurement

Stone for the project will also satisfy the technical requirements given below:

#### Portland Based Limestone

|                        | Guideline Value * |                             | Test method              |
|------------------------|-------------------|-----------------------------|--------------------------|
| Density                | >2100             | kg/m <sup>3</sup>           | BS EN 1936               |
| Porosity               | 10 to 25          | %                           | BS EN 1936               |
| Saturation Coefficient | 0.6 to 0.85       |                             | BRE BR 141               |
| Flexural Strength      | ≥3.4*             | MPa                         | BS EN 12372/ BS EN 13161 |
| Compressive Strength   | ≥28*              | MPa                         | BS EN 1926               |
| Frost resistance       | ≤20               | % loss in flexural strength | BS EN 12371              |

#### White Moleanos Limestone

|                                                                                           | Guideline Value * |                             | Test method              |
|-------------------------------------------------------------------------------------------|-------------------|-----------------------------|--------------------------|
| Density                                                                                   | ≥2160             | kg/m <sup>3</sup>           | BS EN 1936               |
| Porosity                                                                                  | 5 to 15           | %                           | BS EN 1936               |
| Saturation Coefficient                                                                    | 0.8               |                             | BRE BR 141               |
| Flexural Strength<br>To be tested in orientation representative of in-service performance | ≥6.9*             | MPa                         | BS EN 12372/ BS EN 13161 |
| Compressive Strength                                                                      | ≥55*              | MPa                         | BS EN 1926               |
| Frost resistance                                                                          | ≤20               | % loss in flexural strength | BS EN 12371              |

## Beige Moleanos Limestone

|                                                                                                 | Guideline Value * |                                | Test method                 |
|-------------------------------------------------------------------------------------------------|-------------------|--------------------------------|-----------------------------|
| Density                                                                                         | $\geq 2500$       | kg/m <sup>3</sup>              | BS EN 1936                  |
| Porosity                                                                                        | $\leq 10$         | %                              | BS EN 1936                  |
| Flexural Strength<br>To be tested in orientation<br>representative of in-service<br>performance | $\geq 6.9^*$      | MPa                            | BS EN 12372/ BS<br>EN 13161 |
| Compressive Strength                                                                            | $\geq 55^*$       | MPa                            | BS EN 1926                  |
| Frost resistance                                                                                | $\leq 20$         | % loss in flexural<br>strength | BS EN 12371                 |

## Granite

|                      | Guideline Value * |                                | Test method                 |
|----------------------|-------------------|--------------------------------|-----------------------------|
| Density              | $\geq 2560$       | kg/m <sup>3</sup>              | BS EN 1936                  |
| Water absorption     | $\leq 0.4$        | %                              | BS EN 13755                 |
| Flexural Strength    | $\geq 8.3^*$      | MPa                            | BS EN 12372/ BS<br>EN 13161 |
| Compressive Strength | $\geq 131^*$      | MPa                            | BS EN 1926                  |
| Frost resistance     | $\geq 20$         | % loss in flexural<br>strength | BS EN 12371                 |

\*Guideline values normally expected of stones suitable for use as cladding, however the strength of the material will also be sufficient to meet the appointed Façade Contractor's design

### 7.10.2.3 Test data

To demonstrate compliance the declaration of performance (DoP) obtained as part of the CE marking and test data for the tests listed in the above table shall be submitted with the tender. If an alternative stone is proposed for the Works the mechanical and physical properties in the above table may need to be revised by the Engineer. To demonstrate compliance the declaration of performance obtained as part of the CE marking and test data for the tests listed in the above table shall be submitted with the tender.

Test data shall relate to a stone from the same quarry and bed(s) as that which shall be procured and processed; it shall not be more than 18 months old; and it shall be from a laboratory accredited to carry out these tests.

Should recent test data not be available, approval testing specified in 7.10.2.4 shall be undertaken.

#### 7.10.2.4 Stone approval testing (pre-production)

Approval testing is expected to be undertaken before any stone is processed for the Works. The appointed Façade Contractor will submit the results to the Architect.

These tests are required to make an initial appraisal of the suitability of the stones being supplied for this project, and to ascertain its strength and durability characteristics to inform the design of the stone panels. Limestone and Granite samples will be taken from three locations at the working quarry face representing a widely spaced area across the intended extraction zone.

Mean values are to be computed for each group of tested samples. The result of all samples tested within each group is to be used in determining the mean value. If any specimen gives a value of 20% or more below or above the mean, it will be reported and the significance of that result reviewed by the Architect.

All test specimens will be retained after testing to allow further inspection if required.

The following approval tests are required:

##### **Portland Based Limestone/ White Moleanos Limestone**

- a. Density and porosity (BS EN 1936) - 6 No. specimens taken from each block.
- b. Saturation Coefficient (BRE BR 141) – 4 No. specimens taken from each block
- a) Flexural Strength (BS EN 12372/ BS EN 13161) - 10 No. specimens taken from each block. Tested in the dry and wet condition.
- b) Compressive strength (BS EN 1926) - 10 No. specimens taken from each block. Tested in the dry and wet condition.
- c) Frost resistance (BS EN 12371) – Technological Test (Test A) - 10 No. specimens taken from each block.
- d) Petrographic examination (BS EN 12407) – 1 No. specimen

Samples will be tested in the orientation representative of their in-service performance.

##### **Beige Moleanos Limestone**

- c. Density and porosity (BS EN 1936) - 6 No. specimens taken from each block.
- e) Flexural Strength (BS EN 13161) - 10 No. specimens taken from each block. Tested in the dry and wet condition.
- f) Compressive strength (BS EN 1926) - 10 No. specimens taken from each block. Tested in the dry and wet condition.

g) Frost resistance (BS EN 12371) – Technological Test (Test A) - 10 No. specimens taken from each block.

h) Petrographic examination (BS EN 12407) – 1 No. specimen

Samples will be tested in the orientation representative of their in-service performance.

### **Angola Black Granite**

d. Density and porosity (BS EN 1936) - 6 No. specimens taken from each block.

i) Water absorption (BS EN 13755) - 6 No. specimens taken from each block.

j) Flexural Strength (BS EN 13161) - 10 No. specimens taken from each block. Tested in the dry and wet condition.

k) Compressive strength (BS EN 1926) - 10 No. specimens taken from each block. Tested in the dry and wet condition.

l) Frost resistance (BS EN 12371) – Technological Test (Test A) - 10 No. specimens taken from each block.

m) Petrographic examination (BS EN 12407) – 1 No. specimen

Samples will be tested in the orientation representative of their in-service performance

### **7.10.2.5 Production (quality) control testing**

Production control testing will be carried out on successful completion of approval testing.

The appointed Façade Contractor is expected to closely monitor the quality of the stone during production. To this end, an on-going programme of physical testing will be carried out by the previously agreed accredited testing authority.

The blocks of granite shall be tested with the following frequency.

| Production block number | Frequency        |
|-------------------------|------------------|
| 1 to 50                 | every 10th block |
| 51 to 100               | every 20th block |
| 100 +                   | every 50th block |

### **Limestone**

The following production control tests on the Limestone blocks (Portland Basebed, White Moleanos Limestone and Beige Moleanos Limestone):

e. Density and porosity (BS EN 1936) - 6 No. specimens taken from each block.

- f. Visual inspection for cracks – The back face of 10% of the cut to size panels shall be visually inspected for cracks by wetting the back face with water.

Samples will be tested in the orientation representative of their in-service performance.

### **Granite (plinth stone)**

The following production control tests are required:

- g. Density and porosity (BS EN 1936) - 6 No. specimens taken from each block.
- n) Water absorption (BS EN 13755) - 6 No. specimens taken from each block.
- o) Visual inspection for cracks – The back face of 10% of the cut to size panels shall be visually inspected for cracks by wetting the back face with water.

amples will be tested in the orientation representative of their in-service performance.

The stones will be deemed to satisfy the performance requirements if, in the opinion of the Architect, the results of the production control testing demonstrate;

- h. Consistency with the approval testing results.
- i. Technical performance in excess of the requirements given in this Specification.

The appointed Façade Contractor will reject all the stones represented by specimens which fail to satisfy the performance requirements.

Production control testing will be carried out on selected blocks of the chosen stone.

### **7.10.3 Range Samples**

As soon as possible after award of Contract, the appointed Façade Contractor will arrange for a sufficient number slabs of each of the approved stone types to be cut from at least three randomly selected blocks taken from current workings at the quarry, from which the visual range for the project can be defined. Sufficient slabs will be provided to include all the features which, in the opinion of the stone supplier, are likely to occur in the stone and therefore should be considered in determining the acceptable visual range. These slabs will then have the specified finish applied and will be set up in the quarry with a neutral background and orientated facing east or west to enable viewing in shadow or sunlight. Features not represented on these slabs will not be included in stone supplied for the project.

The Architect will then visit the quarry, view the finished slabs and agree an acceptable range of colour, veining, blemishes and other relevant visual criteria with the appointed Facade Contractor. These samples will also be viewed both dry and then wet. Features considered technically or visually unacceptable will be clearly marked on the slabs. At the

same time the Architect and the appointed Façade Contractor will agree the steps that will be taken to ensure the correct tonal distribution of stones across the building.

The appointed Contractor will then arrange for 3 sets of range samples, for each of the agreed stone types, to be cut from the approved slabs or provide 3 sets of duplicate samples. The range samples will be distributed between the stone supplier, the Architect and appointed Contractor for future reference. The range samples should include all relevant features and characteristics illustrated in the agreed range. Records of unacceptable features should be kept by the stone supplier and appointed Contractor.

The appointed Contractor will use these range samples to assess the visual quality of the stones processed during the fabrication and installation of the Cladding. He will immediately notify the Architect of any variations outside the agreed range.

#### 7.10.4 Requirements at the quarry

All stone for each stone type will be cut from a designated part of the quarry that exhibits consistent mechanical properties.

The extraction of the stone will be undertaken by experienced and competent quarrymen and they will use only those techniques and practices which will not impair the performance of the finished product.

Each block will be given a unique reference number and marked to show the part of the quarry / mine from which it is worked, the date of extraction, the volume of the block (as defined in BS EN 1467) and the direction of bedding or rift.

Should the character of the stone change during the progress of extraction from the quarry, the appointed Façade Contractor will immediately inform the Architect so that an alternative location of working can be agreed. Under no circumstances will the appointed Contractor obtain stone from another quarry or another part of the nominated quarry without the prior agreement of the Architect.

In the case of stone that requires ‘seasoning’ the appointed Contractor will ensure that sufficient time has been allowed under suitable storage conditions for the stone to achieve maximum strength and durability.

### 7.11 Glazed Extruded Terracotta

The terracotta is expected to be a glazed, frost resistant material suitable for exterior application. The terracotta will be of uniform texture free from voids or inclusions. It will have the following properties, as determined by the BS EN 771-1 and BS EN ISO 1545:

| Number            | Title                                                                             |
|-------------------|-----------------------------------------------------------------------------------|
| BS EN ISO 10545-1 | See 9.3.1                                                                         |
| BS EN ISO 10545-2 | See 8.5                                                                           |
| BS EN ISO 10545-3 | Guideline values:<br>Water absorption; < 10%<br>Density: >2,000 kg/m <sup>3</sup> |
| BS EN ISO 10545-4 | > 4Mpa*                                                                           |

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| BS EN ISO 10545-5  | TBC                                                             |
| BS EN ISO 10545-8  | No damage                                                       |
| BS EN ISO 10545-9  | No damage                                                       |
| BS EN ISO 10545-10 | State results                                                   |
| BS EN ISO 10545-11 | No visible crazing at 1 metre viewing distance in diffuse light |
| BS EN ISO 10545-12 | No damage                                                       |
| BS EN ISO 10545-13 | No damage                                                       |
| BS EN ISO 10545-14 | None visible after cleaning                                     |
| BS EN ISO 10545-15 | No off gassing                                                  |
| BS EN ISO 10545-16 | Within ranges determined in section 6.3                         |
| BS EN ISO 771-1    | S2 (low)                                                        |

\* Guideline values normally expected of terracotta suitable for use as cladding, however the strength of the material will also be sufficient to meet the appointed Façade Contractor's design.

## 7.12 Mortars

Mortars will be lime mortars graded to provide sufficient flexibility to accommodate the movements and joint size tolerances expected of the façade, whilst providing a sufficient level of frost resistance and durability for the building location. Examples include lime putties (non hydraulic lime and sharp sand) or NHL2 mortars (natural hydraulic lime) containing a good range of particle sizes.

Mortar ingredients will conform to the following standards and will be approved via samples by the Architect:

- Natural Hydraulic Lime (NHL) to BS EN 459: Part 1: 2001;
- Calcium Lime (CL90) to BS EN 459: Part 1: 2001;
- Sand to BS EN 13139: 2002;
- Pigments to BS EN 12874;

Mortars for stone shall have a minimum compressive strength of 3.5N/mm<sup>2</sup>.

It shall be noted that the strength of mortars should be varied in different areas of the façade. For example, exposed areas such as cappings may require a stronger mortar to provide greater durability.

Mortar colour shall match the Architect's design intent, and where pigments are required to achieve the desired colour, these shall be stable and unaffected by light and lime.

### 7.12.1 Hydraulic lime mortar

Hydraulic lime mortars shall typically achieve a minimum mean compressive strength of 3.5N/mm<sup>2</sup> at 91 days. Lime mortars shall be graded to provide sufficient flexibility to accommodate the movements and joint size tolerances expected of the façade, whilst

providing a sufficient level of strength, frost resistance and durability for the building location.

Feebly, moderately and eminently natural hydraulic lime (NHL) mortars shall comply with BS EN 459-1: 2010. These may be site-mixed or supplied pre-packaged.

### **7.12.2 Non-hydraulic lime mortars and lime putty**

Lime shall consist of high calcium or dolomitic lime and be supplied from an approved source. Calcium lime shall comply with BS EN 459-1: 2010. Lime putty shall be stored in airtight bins.

### **7.12.3 Aggregate for mortars**

Well graded sands and aggregates shall comply with BS EN 13139: 2002.

### **7.12.4 Pozzolanic additives**

Pozzolanic additives such as pulverised fuel ash (PFA) or high temperature insulation (HTI) powder may be added to lime mortar to improve frost resistance. These shall be graded and assessed in accordance with BS 8221-2: 2000.

### **7.12.5 Pigments**

Pigments shall comply with BS EN 12874.

### **7.12.6 Hydraulic lime mortar testing**

#### **7.12.6.1 Preliminary testing**

Sample brickwork mortars shall be tested for mean compressive strength at least six weeks prior to construction of the stonework. Half of the samples shall be tested at 28 days after curing, and the remainder shall be tested at 91 days to ensure full strength has been achieved. Retarded mixes shall have extended curing periods to include retardation period.

Testing shall be undertaken by a UKAS accredited laboratory.

Mortars shall be tested to the following methods:

Site batched and mixed mortar: test to BS 5628-1 Appendix A1 and BS 4551-1. (40 x 40 x 100 mm prisms).

Mortars manufactured to BS EN 998-2: test to BS EN 1015-11. (40 x 40 x 160 mm prisms)

Test results shall be submitted to the Contract Administrator.

#### **7.12.6.2 Site testing**

Six specimens per 150m<sup>2</sup> of walling or six specimens per storey, whichever the more frequent shall be tested for compressive strength. Specimens shall be as preliminary tests,

but prepared during construction. Again, half of the samples shall be tested at 28 days after curing, and the remainder shall be tested at 91 days to ensure full strength has been achieved. Retarded mixes shall have extended curing periods to include retardation period.

Test results shall be submitted to the Contract Administrator.

## 7.13 Gaskets

Gaskets and seals used to achieve the required weather and airtightness shall be selected in accordance with BS EN 12365-1 to fully accommodate the range of dimensional tolerances associated with fabrication and installation of the Works. They shall be formed with materials capable of maintaining their elastic qualities, dimensions and resistance to physical and chemical attack sufficient to maintain the full performance during the design life. Gaskets providing the water and air seals shall be formed into complete frames with factory formed injection moulded vulcanised corner joints.

Gaskets shall be free from contact and migration stain and shall be compatible with all substrate, sealant and finishes with which they are likely to come into contact. The gaskets shall be free of mould flash.

Extruded rubber gaskets shall comply with BS 4255. Cellular profiled gaskets shall comply with ASTM-C509. The Sub-contractor shall comply with the manufacturer's recommendations for specific conditions of use wherever a range of hardness or compressibility is available for a product. All materials must be compatible with substrate, sealants, finishes and other materials in joint systems.

## 7.14 Insulation

### 7.14.1 General

Thermal insulation shall be inert, durable, rot and vermin proof, CFC and HCFC free, shall not support mould fungal or bacteria growth and shall provide the specified performance for the service life of the Works. Due allowance shall be made for the reduced thermal performance of insulation due to the effects of moisture and ageing.

Thermal insulation shall be sufficiently robust and cohesive to allow it to be removed and replaced in case of necessary adjustment during installation without loss of material or performance. It shall not be injurious to human health when being properly installed, when in service and when being removed or replaced.

Insulation shall be Mineral wool to BS EN 13162.

Where rigid polyurethane foam or phenolic foam is used, they shall be used in conjunction with adequate fire barriers installed to prevent fire spreading within the cladding system and any fire from coming into direct contact with them.

### 7.14.2 Embodied energy

Insulation shall be selected to achieve an A+, or A rating as defined in the Green Guide to Specification.

### 7.14.3 Responsible sourcing of insulation products

At least 80% of the insulation used in the Works shall be responsibly sourced to either BES 6001 or BS 8902; or must be from a source with a certified Environmental Management System (EMS) for product manufacture and key supply chain processes. Independently certified EMS schemes include BS EN ISO 14001, or equivalent.

A copy of the BES 6001 certificate number, and/or EMS certificate shall be provided to demonstrate how responsible sourcing and/or environmental management shall be implemented from Award of Contract through to Contract Completion.

## 7.15 Setting Blocks

Setting blocks shall be polychloroprene (Neoprene), polypropylene or EPDM of an appropriate Shore A hardness. Carbon compounds for fillers and durability agents shall not leach out over time and cause staining.

## 7.16 Sealant (non-structural)

In general, sealant shall be selected and applied in accordance with the guidelines contained in BS 6213, BS EN ISO 11600 and CIRIA Guide 178. Sealant shall be:

- Silicone sealant.
- One or two part polysulphide sealant.
- As proposed by the Sub-contractor to the appropriate British or European Standard, subject to written confirmation being provided by the manufacturer that his product is being used in appropriate manner in an appropriate situation.

Backing rods shall be closed cell polyethylene foam. When closed cell sections are used non-gassed produced shall be selected.

The colour of visible sealant used shall be as specified on the Architects Drawings.

## 7.17 Materials used for Flashings and Vapour Control Layers

Flashings and vapour control layers shall maintain their performance and properties for the expected service life of the Works. They shall resist the deleterious effects of water, cleaning agents, temperature variations expected from the specified temperature ranges, gaseous pollutants (including ozone), weak acids deriving from gaseous pollutants dissolved in water and UV radiation to which they may be exposed during installation and in service.

Flashings shall have appropriate water vapour permeability. The water transmission properties of the material and the complete construction shall preclude the inward passage of liquid water, but permit the release to outside over time of harmful condensation and other trapped water.

Flashings shall have the mechanical properties to withstand installation and specified design loads.

The material used shall be lead, stainless steel, coated aluminium or one of the following membranes:

- Flexible ethylene-propylene diene monomer (EPDM)
- Flexible chlorosulphonated polyethylene
- Flexible chloroprene. Non-cellular chloroprene products shall comply with the requirements of BS 4255-1
- Flexible butyl rubber

Flexible membranes shall meet the requirements of BS EN 13984.

Vapour control layers shall have a minimum vapour resistance, of 200 MN.s/g. The vapour resistance (BS EN ISO13788+ BS 5250) of the material shall preclude the passage of water vapour from the internal to the cold side of the construction in order to avoid the formation of interstitial condensation. Any water vapour already in the construction shall be allowed to evaporate to the outside.

Performance shall be maintained at joints between the individual pieces of the vapour control layer material.

Proprietary membrane waterproofing systems shall be CE marked and shall have product data sheets giving details of performance and verification testing. The Sub-contractor shall provide sufficient technical information to confirm that the system proposed meets the requirements of this specification.

## **7.18 Breather membranes**

Breather membrane shall be Type 1 as defined in BS 4016.

## **7.19 Timber**

### **7.19.1 Sources**

Timber will be sourced from managed approved sustainable forests that are certified by an International Assessing Agency recognised by the Forest Stewardship Council (FSC) and will have a full documentation proving that the material supplied and installed is from sustainable sources. No timber will be brought to site that is not traceable and subject to the relevant FSC certification. All suppliers will have the appropriate FSC accreditation.

## 7.20 Render

### 7.20.1 General

The selected render system shall be a proprietary render system that holds a suitable certificate of conformity for use in this application.

Render samples indicating the range and appearance of the final works shall be submitted to the Employer's Agent for approval prior to ordering.

### 7.20.2 Visual appearance

The rendered finish should be applied to the surface in a uniform layer to the thickness specified.

Continuous surfaces should be completed without a break.

Care should be taken in the detailing of the system around openings and projections.

Where indicated on the Architects drawings, movement and architectural joints should be incorporated utilising details to manufacturer's recommendations.

### 7.20.3 Technical requirements

The render system shall comply with BS EN 13914-1

The system shall comprise the following components;

- Insulation
- Carrier board
- Basecoat
- Reinforcement
- Primer
- Finishing coats

## 8 Finishes

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### 8.1 Anodised Aluminium

#### 8.1.1 Generally

Anodising will be in accordance with BS 3987.

Anodic oxidation of wrought aluminium shall comply with the minimum requirements of BS 3987.

The anodising shall be carried out by an Anodiser nominated by the Sub-contractor and shall be subject to the agreement of the Architect. The same Anodiser shall be used for the duration of the Works. The Anodiser shall operate an ISO 9001 approved quality assurance system.

#### 8.1.2 Materials

Materials shall comply with Appendix C of BS 3987 and other performance criteria detailed in this specification.

The Sub-contractor shall provide details of the chosen alloys to the Architect for approval before procuring any materials.

#### 8.1.3 Significant surfaces

All visible surfaces and other surfaces exposed to the external environmental conditions shall be deemed to be significant surfaces. These shall include any drained and ventilated cavities and pressure equalisation chambers.

In so far as is technically feasible, the positioning of die bridges for hollow section profiles shall be such that the impact of 'weld lines' shall be minimised on significant surfaces. In any event, the extent of weld and die lines shall be subject to agreement through sample approval, refer to.....below.

The Sub-contractor shall note all significant surfaces on his drawings and provide the Architect with written confirmation from the Anodiser that full coverage in accordance with this specification is achieved.

#### 8.1.4 Fabrication requirements

Fabrications may be from pre-finished lineals or machine lengths at the recommendation of the Anodiser. Care shall be taken to avoid scratching and damage to the anodic coatings during the fabrication. Where machining could leave exposed un-anodised surfaces, e.g. pivot hung ventilators, etc or visible cut edges on sheet metalwork, the work shall be anodised after all machining. Where anodising is to be carried out after machining, satisfactory jigging points shall be agreed between the Sub-contractor and the Anodiser and then the Architect. Two sample joints shall be submitted to the Architect for review, prior to fabrication.

### 8.1.5 Anodising conditions

Anodising shall be carried out under conditions of acceptable good practice for architectural applications. The processing conditions shall be held constant for the duration of the period required for completion of the contract. Records of these conditions shall be kept and made available for inspection in the event of a dispute as to the quality of the finished anodic coating.

### 8.1.6 Anodising process

The anodic finish colour shall be as specified by the Architect or as noted on the Architects drawings.

### 8.1.7 Colour control limits for natural anodising

The following colour limits shall apply:

| Element            | Maximum permitted colour variation, $\Delta E$ (CIELAB formula according to ISO 7724/3) |
|--------------------|-----------------------------------------------------------------------------------------|
| Aluminium profiles | 3                                                                                       |
| Sheet/ plate       | 2                                                                                       |

Once production has commenced colour control limits outside of the above range shall be subject to the approval of the Architect.

### 8.1.8 Samples

#### 8.1.8.1 General

Approval to commence manufacture shall only be given subject to the provision and acceptance of the samples detailed below.

#### 8.1.8.2 Natural anodising

Prior to commencement of the anodic coating, the Contractor, in conjunction with the Anodiser, shall submit to the Architect for review, the following samples showing the expected occurrence of die and weld lines.

- a. 2 No hollow section extrusions, typical for the Works, each 1200mm long;

Once agreed and signed off by the Architect, the Contractor shall label and sign each sample in five separate locations and arrange for the samples to be cut into five equal sized pieces. The sample pieces shall then be distributed as follows:

- a. The Contractor;
- b. The Architect (2 pieces);

- c. The Anodiser;
- d. The Independent Inspector;

The respective parties shall keep them for one year after Practical Completion. The Architect shall use these samples to agree or otherwise any disputed finishes.

### 8.1.8.3 Coloured anodising

Prior to commencement of the anodic coating, the Sub-contractor, in conjunction with the Anodiser, shall submit to the Architect for review, the following samples showing the variations in colour and texture and the expected occurrence of die and weld lines.

- a. 2No.extrusions, typical for the Works, each 1200mm long showing the upper and lower range limits;
- b. 2No. sheets, typical for the Works, but each not less than 1200mm x 200mm, showing the upper and lower range limits;

Once agreed and signed off by the Architect, the Sub-contractor shall label and sign each sample in five separate locations and arrange for the samples to be cut into five equal sized pieces. The sample pieces shall then be distributed as follows:

- a. The Contractor;
- b. The Architect ( 2 pieces);
- c. The Anodiser;
- d. The Independent Inspector;

The respective parties shall keep them for one year after Practical Completion. The Architect shall use these samples to agree or otherwise any disputed finishes.

### 8.1.9 Acceptance testing

#### 8.1.9.1 Third party inspection

The Sub-Contractor shall make provision for the cost of a minimum of three acceptance inspections to be conducted by an Independent Inspection Authority. The Independent Inspection Authority shall be experienced in the assessment of anodised aluminium finishes. The Sub-contractor shall provide details of the proposed Independent Inspection Authority with his Tender.

If, for whatever reason, third party inspection is not implemented as specified, the Architect shall instruct an Independent Inspection Authority to carry out the specified inspections, and shall make an adjustment to the Sub-contract sum to cover the costs of the inspections.

In the case of dispute between the Anodiser's results and those of the Independent Inspection Authority, the results of the Independent Inspection Authority shall take precedence.

### 8.1.9.2 Sampling procedure

Sampling procedures and plans shall comply with BS 6001: Part 1 (ISO 2859 Part 1), for general inspection Level 2, AQL = 1%. These inspections shall be carried out at the finishing plant prior to fabrication

### 8.1.9.3 Visual appearance

The visual appearance of anodised aluminium shall comply with the requirements of BS 3987, except that the viewing distance for both external and internal architectural applications shall be 1000mm unless otherwise agreed.

### 8.1.9.4 Sealing quality

Sealing of the anodic coating shall comply with BS 3987. Cold impregnation sealing processes shall not be used.

For production control purposes the sealing quality shall be determined using BS EN 12373-5. The corrected admittance value (i.e. the value it would be at 25°C and at a thickness equivalent to 20µm) should not exceed 25µs.

In cases of dispute the method described in BS 3987 shall be used as the referee method. A minimum of 2 No.samples with the highest admittance values shall be selected at random and tested.

### 8.1.9.5 Surface abrasion resistance

Surface abrasion resistance shall comply with BS 3987. In cases of dispute, the surface abrasion resistance shall be assessed using the method described in BS EN ISO 12373: Part 9. A minimum of 8 No.samples shall be selected at random from the batch of material suspected of low surface abrasion resistance. Anodic oxide films produced in sulphuric acid electrolytes shall have a wear index of not greater than 1.4.

### 8.1.9.6 Certification

Inspection certificates demonstrating that all relevant tests required by BS 3987 and this specification have been carried out and that the anodic finish complies with the minimum requirements in all respects, shall be submitted on request, to the Architect for review.

### 8.1.10 Temporary protection

All coated surfaces vulnerable to damage during handling and installation or by subsequent site operations, shall be fully protected for the duration of the Works.

Protective coverings shall be resistant to all weathers. Prior to installation, they shall be removed from areas inaccessible after installation. Where necessary, they shall be partially removable and replaceable for access to fixing points during installation and/or subsequent site operations.

Any protective tapes used in direct contact with the coating shall be low tack, self-adhesive type in white or any colour lighter than the coating to be covered. It shall be applied and removed in accordance with the recommendations of the Anodiser's guidelines.

Protective tapes shall not be kept in contact with coated surfaces for longer than 6 months.

Should the protective covering need to remain in place following installation, the Sub-contractor, shall, with the Architect, agree a programme for monthly inspections and make good/replace the coverings as may be required.

#### **8.1.11 Damaged anodised aluminium finishes**

The rectification of damage to anodised aluminium finishes located internally shall be permitted subject to an agreed repair benchmark and method.

The rectification of damaged anodised aluminium finishes located externally shall not be permitted. Any components with damaged anodised finishes shall be replaced with new.

#### **8.1.12 Maintenance**

The Sub-contractor shall provide comprehensive instructions for the maintenance of the coating. He shall ensure that the Anodiser has endorsed the recommendations and that, when applied as specified, they fulfil the requirements of the coating Warranty.

The recommendations shall include, but not be limited to the following:

- Frequency of washing
- Solutions/detergents to be used and recommended dilution levels
- Type of cleaning materials
- Recommendations (if any) for deep cleaning processes (including solutions/compounds etc.) and their frequency

#### **8.1.13 Polyester powder coated Aluminium**

##### **8.1.14 General**

Polyester powder coating on aluminium shall comply with the minimum requirements of "The specifications for a quality label for paint, lacquer and powder coatings on aluminium for architectural applications" – Qualicoat Specifications 12th Edition.

The coating shall be carried out by an applicator nominated by the Trade Contractor, approved by the polyester powder manufacturer and agreed with the Architect. The same powder coating applicator shall be used for the duration of the Works.

##### **8.1.1 Powder**

A Qualicoat Class 1 powder shall be used.

### 8.1.2 Service life

The service life shall be as defined in the Durability section of this Specification.

### 8.1.3 Pre-treatment

A chromate pre-treatment shall be used. Alternative pre-treatments shall not be used unless approved by the Architect.

### 8.1.4 Colour control

Colour control samples shall be submitted with the Tender.

Prior to commencement of the coating, the Sub-contractor, in conjunction with the Powder Coater, shall submit to the Architect for review, the following samples showing the variations in colour and texture:

- 2No. extrusions, typical for the Works, each 1200mm long showing the upper and lower range limits.
- 2No. sheets, typical for the Works, but each not less than 1200mm x 200mm, showing the upper and lower range limits.

Once agreed and signed off by the Architect, the Sub-contractor shall label and sign each sample in five separate locations and arrange for the samples to be cut into five equal sized pieces. The sample pieces shall then be distributed as follows:

- The Sub-contractor
- The Architect (2 pieces)
- The Powder Coater
- The Independent Inspector

The respective parties shall keep them for one year after Practical Completion. The Architect shall use these samples to agree or otherwise any disputed finishes.

### 8.1.5 Third party inspection

The Sub-contractor shall make provision for the cost of a minimum of three acceptance inspections to be conducted by an independent inspection authority. The independent inspection authority shall be experienced in the assessment of powder coated aluminium finishes. The Sub-contractor shall provide details of the proposed independent inspection authority at the time of tender to the Architect for review.

If, for whatever reason, third party inspection is not implemented as specified, the Architect shall instruct an independent inspection authority to carry out the specified inspections and shall make a financial adjustment to the Sub-contract sum to cover the costs involved.

In the case of dispute between the powder coating applicator's test results and those of the independent inspection authority, the results of the independent inspection authority shall take precedence.

#### **8.1.6 Sampling procedures**

Sampling procedures and plans shall comply with BS 6001: Part 1 (ISO2859/2) for general inspection Level 2, AGL = 1% on each colour and finish used in the Works. These inspections must be carried out in the powder coating plant prior to fabrication.

#### **8.1.7 Testing requirements**

Unless specified otherwise, all inspections and tests on the finished coating shall comply with the minimum requirements of Qualicoat Specification.

#### **8.1.8 Applicators certificate**

The Sub-contractor shall provide the Architect with certificates which state that the finish complies with the requirements of this specification.

#### **8.1.9 Damaged polyester powder coat finishes**

The rectification of damage to polyester powder coat finishes shall not be permitted. Any components with damaged polyester powder coat shall be replaced with new.

#### **8.1.10 Temporary protection**

All coated surfaces vulnerable to damage during handling and installation or by subsequent site operations shall be fully protected for the duration of the works.

Protective coverings shall be resistant to all weathers and be removed from areas inaccessible after installation. They shall be partially removable and replaceable for access to fixing points during installation and/or subsequent site operations.

Any protective tapes used in direct contact with the coating shall be a low tack, self-adhesive type in white or any colour lighter than the coating to be covered. It shall be applied and removed in accordance with the recommendations of the powder manufacturer guidelines.

Protective tapes shall not be kept in contact with coated surfaces for longer than 6 months.

### **8.2 Steelwork Corrosion Protection - Galvanising**

Mild steel components shall be hot-dip galvanised after forming and drilling in accordance with BS EN ISO 1461. The cutting, drilling or working of galvanised components shall not be permitted.

The galvaniser shall control dipping rates to ensure distortion is minimised, and cracking due to differential expansion is avoided.

All galvanised components shall be subject to the following inspections to ensure freedom from cracking:

- 100% visual inspection
- Magnetic particle inspection of areas indicated on the Architects drawings

All inspections shall be recorded

Any cracked components shall be rejected.

Cut ends or surface working to pre-galvanised sheet steel or any damage to galvanising (hot-dipped or otherwise) shall be repaired with zinc rich paint complying with BS 4652.

## **8.3 Steelwork Corrosion Protection – Paint**

### **8.3.1 General**

Painting shall follow the requirements of The National Structural Steelwork Specification for Building Construction (NSSSBC) 5th Edition, Section 10, Protective Treatment. The following clauses in section 10 are not required:

- Surface preparation - clause 10.2.1.
- Sprayed metal coatings - clause 10.3.

Painting shall be carried out in accordance with the Paint Manufacturer's instructions and section 10 of the NSSSBC. The painting shall be carried out by an Applicator approved by the Paint Manufacturer and reviewed by the Architect. The same Applicator shall be used for the duration of the Works. The Sub-contractor shall also obtain from the Applicator a detailed method statement for the preparation and coating process and submit it to the Architect for review.

### **8.3.2 Performance Criteria**

The colours and gloss levels shall be as specified on the Architects Drawings.

The finished coating shall be maintenance free with the exception of regular cleaning as prescribed, be durable and have a colourfast life of at least 15 years. The Sub-contractor shall collate comprehensive maintenance requirements from the Paint Manufacturer and the Applicator for inclusion in the Operations and Maintenance Manual.

The Sub-contractor shall note the anticipated surface temperatures and environmental conditions described herein, and shall obtain written confirmation from the Paint Manufacturer that the system as specified will comply with the requirements.

### **8.3.3 Application**

All visible surfaces and other surfaces exposed to the external environment conditions shall be painted. These shall include any drainage and ventilated cavities.

The minimum dry film thicknesses (DFT) shall be those quoted herein.

The finished surface shall be as per the control samples submitted to the Architect.

All materials shall comply with the max VOC given in the current edition of the Secretary of State's Process Guidance Note PG 6/23.

Zinc-rich primers shall comply with BS 4652.

### 8.3.4 Control Samples

Colour control samples shall be submitted with the Tender.

Prior to commencement of the coating, the Sub-contractor in conjunction with the Applicator shall submit to the Architect for review, the following samples showing variations in colour, texture and gloss.

- 2 No rolled steel sections 1200mm long showing the upper and lower range limits.
- 2 No folded sheet sections 1200mm x 200mm showing the upper and lower range limits.

Once agreed and signed off by the Architect, the Sub-contractor shall label and sign each sample in four separate locations and arrange for the samples to be cut into four equal sizes. The samples shall then be distributed as follows:

- The Sub-contractor
- The Architect (2 No. samples)
- The Applicator

The respective parties shall keep them for one year after Practical Completion. The Architect shall use these samples to agree or otherwise any disputed finishes in the Works.

### 8.3.5 Remedial Works

Repairs and remedial works shall be carried out in accordance with the original specification, agreed with the Architect.

The Sub-contractor shall submit to the Architect for review, a detailed method statement for making good damaged areas of coating. Details are to include surface preparation of both exposed steel and other coatings, choice of materials if these differ from those originally used, means of application, conditions of application and any other relevant considerations.

All remedial works shall be subject to trials and rejections shall be at the discretion of the Architect. The Architect shall keep the agreed trial sample as a control sample. Repair record sheets shall be logged by the Sub-contractor and placed in the Operations and Maintenance Manual as a record of the as-built works.

### 8.3.6 External Environment Paint System

|                     |                                                                           |
|---------------------|---------------------------------------------------------------------------|
| Surface Preparation | Blast clean to Sa2½ of BS 7079 Part A1                                    |
| Primer Coat         | Zinc-rich epoxy to minimum DFT 75 µm. Shop applied                        |
| Intermediate Coat   | Epoxy micaceous iron oxide (MIO) to minimum DFT 100-125 µm. Shop applied. |
| Finish              | 2 coats of Acrylic / Urethane to minimum DFT 50 µm. Shop/site applied.    |

### 8.3.7 Internal Environmental Paint System

|                     |                                                                                      |
|---------------------|--------------------------------------------------------------------------------------|
| Surface Preparation | Blast clean to Sa2½ of BS 7079 Part A1.                                              |
| Primer Coat         | Epoxy zinc phosphate to minimum DFT 50 µm. Shop applied.                             |
| Finish              | As required by the Architect and recommended by the manufacturer. Shop/site applied. |

## 8.4 Steelwork Corrosion Protection - Galvanising

Mild steel components shall be hot-dip galvanised after forming and drilling in accordance with BS EN ISO 1461. The cutting, drilling or working of galvanised components shall not be permitted.

The galvaniser shall control dipping rates to ensure distortion is minimised, and cracking due to differential expansion is avoided.

All galvanised components shall be subject to the following inspections to ensure freedom from cracking:

- 100% visual inspection
- Magnetic particle inspection of areas indicated on the Architects drawings

All inspections shall be recorded

Any cracked components shall be rejected.

Cut ends or surface working to pre-galvanised sheet steel or any damage to galvanising (hot-dipped or otherwise) shall be repaired with zinc rich paint complying with BS 4652.

## 8.5 Bronze Finishes – Significant Surfaces

### 8.5.1 General

Unless otherwise indicated all visible surfaces are considered significant surfaces and will be subject to the specified finishing treatment.

### 8.5.2 Patination

The copper and copper alloys will be subject to a patination treatment to achieve the desired coloured finish.

The appointed Contractor will submit a patination method statement accompanying each different combination of copper alloy and manufacturing process for approval by the Architect.

The appointed Contractor is expected to ensure adequate protection measures are employed to prevent damage to other materials from spillage and/or cross-contamination from excess patination chemicals.

The patination process and chemical mix will not remain active upon installation of the bronze work.

### 8.5.3 Colour

The final finish colour will be as specified by the Architects and via approved samples.

## 8.6 Extruded glazed terracotta

### 8.6.1 General

Terracotta will be manufactured in accordance with the manufacturer's quality assurance programme and internal codes of practice, in addition to the standards specified in this document.

The body of the tile will be made from ground, fine quality Westerwald clays extruded with horizontal perforations. The body will be mixed with ground, previously fired fireclay materials (chamotte) to control shrinkage, stability and provide surface texture.

The external finish will be conventional silica based ceramic glaze with metal oxide pigments to produce various colours. Organic or non-conventional glaze systems will not be used.

Refer to the Employers Architect's drawings for further details of colours and proposed glaze by Christine Jetten Studio.

Glazes will be smooth and evenly coat the surface and will be post applied to all cut tile ends and the ends of all standard length units.

No unglazed edges will be visible in the works including vertical cuts.

The appointed Facade-Contractor will ensure that the installed product complies with the Employers Architect's colour choice, is within the acceptable colour variation limits, degree of crazing and texture requirements as detailed elsewhere in this document.

Glaze is to remain intact for the lifetime of the cladding. Chips, cracks and crazing in excess of those described in section 11.5 will not be acceptable.

The number and types of coats of glaze will be in accordance with the manufacturer's specification for the particular glaze systems used. All components will have their significant surfaces coated with the glaze. Glaze thickness and corner arris details will be proposed by the manufacturer for the Employer to review.

The cured film will be dense and consistent, with no seeding and will be free from flow lines, streaks, pin holes, blisters, tear damage and other coating defects/surface imperfections when viewed from a distance of 1 metre under normal diffuse daylight.

No crazing will be visible when viewed from a distance of 1 metre under normal diffused daylight.

### 8.6.2 Control Samples

Prior to commencement of manufacture of the Works, the appointed Facade-Contractor in conjunction with the glazed terracotta manufacturer will make available for inspection at their premises, the following samples, (further samples may be required if those submitted are not acceptable to the Architect):

- a) 3 number of each type of panel from cladding type 10.9;
- b) 3 number of each type of panel from cladding type 11.x;

All sample dimensions will be as the final works, showing the agreed maximum variations in crazing, colour, spot flaws, texture and gloss. Samples to incorporate glaze to any factory cut surface and corners.

Subject to agreement by the Architect, the appointed Facade-Contractor will label and sign selected samples that adequately illustrate the agreed maximum variations in crazing, colour, texture and gloss in four separate locations and arrange for the samples to be cut into four equal sizes. Where it is not practical to cut samples into 4 equal parts, duplicates will be provided. The samples will then be distributed as follows:

- 1) The appointed Facade-Contractor;
- 2) The Architect;
- 3) The terracotta manufacturer;

These will be kept by the respective parties for at least the duration of the Works. The Architect will use these samples to agree or otherwise any disputable finishes in the works.

Any samples used to illustrate the extent of glazing must have first undergone testing to EN ISO 10545-11, as required elsewhere in this specification, to illustrate the possible long term crazing of the panels. These samples can be the same samples used during those tests.

The Architect will carry out an inspection on the first batch of glazed terracotta panels produced and agree with the Manufacturer and appointed Façade Contractor an acceptable range and limit of variation from the control samples within reason.

### 8.6.3 Technical Procedures

The following procedures will be applicable. Manufacturer's method statements and internal codes of practice to be submitted at tender stage:

- a) Clay selection
- b) Clay and aggregate mixing
- c) Glaze selection
- d) Glaze mixing
- e) Transfer production
- f) Glaze colour control and matching
- g) Process testing
- h) Workmanship
- i) Glaze adhesion
- j) Dimensional monitoring
- k) Manufacture QA/QC to ISO9000 where appropriate
- l) Batch mixing

The Manufacturer is expected to exercise due control on the temperatures of the kilns to ensure the correct temperatures and will comply with the procedures as specified by the glaze manufacturer. Records and test certificates incorporating factory temperature, ambient humidity and barometric pressure will be kept of the processing conditions employed and be available for inspection by the Architect at any time.

## 8.7 Render

Final finish is subject to confirmation and approval through process of samples.

Range samples are to be provided with the range of finish that will be achieved for approval by the Employer's Agent.

## 9 Quality Control

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### 9.1 Quality Plan

This section sets out the minimum requirements for the production of a Quality Plan for the Works. Much of the necessary documentation should already exist as part of the Sub-contractor's standard procedures and shall be supplemented where necessary by project specific procedures.

As required by the Schedule of Submissions an outline Quality Plan and a project specific Quality Plan shall be produced by the Sub-contractor following the principles and guidelines of BS EN ISO 9001 to demonstrate how quality management will be implemented from award of Contract through to Contract Completion.

### 9.2 Quality Control Procedures

The Sub-contractor shall develop inspection procedures, check lists and audits for monitoring the Works at the various stages.

A clear system shall be implemented to track information flow and check that work is being designed and installed to the most up-to-date revisions of specifications, drawings, interface drawings and this Contract Document.

The documented procedures shall cover but shall not necessarily be limited to the following:

- Names and CV's of the Quality Manager and his team
- Quality assessment
- List of design development drawings required and the rate at which they will be produced.
- Inspection and test procedures to be adopted in checking the Works
- Stages at which checklists will be used and samples of those checklists
- Statements on works procedures on the correct use of materials and components, both off site and on site.
- List of product information with latest revisions.
- Trade contractors appointed by the Sub-contractor that are involved in the Works.
- The trade contractor's quality systems and details of their checklists.
- Design, procurement, material checking, fabrication, storage, handling, transport and protection requirements.
- Procedure for checking compliance with the Contract Documents and with accepted good practice.
- Procedure for registering and reporting non-compliance.

- Procedure for ensuring that non-compliance does not recur.
- Calibration records of measuring equipment.
- Certification that the Works are considered by the Sub-contractor to be complete and are ready for inspection by the Architect.
- Checklist register to ensure that all items have been inspected by the Sub-contractor and non-compliance discharged.
- Protection of work in progress and work completed, including primary brackets, seals and fire stops.
- Compliance with the requirements of the CDM Regulations.

During fabrication, checklists shall include inspection requirements for all components and finishes that cannot be inspected at later stages in manufacture, assembly, or erection of components.

During erection, checklists shall include such items as survey spot checks of primary brackets, condition of components and assemblies and finishes, joint cleaning, backer rod and sealant installation, and fire stopping.

Completed quality checklists shall accompany the components at all stages of production up to the final erection on site and be available for inspection at any stage.

The Sub-contractor shall keep full records of all quality management documents for a period not less than twelve years from Practical Completion. Copies of the quality management documents shall be included in the Operations and Maintenance manual as a record of the as-built works.

Detailed method statements as referred to in this Contract Document shall be required before the work commences.

## 10 Fabrication

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### 10.1 General

The Works shall be fabricated in accordance with the Sub-contractor's drawings and method statements, modified as necessary to meet the requirements of the Contract Documents. The Sub-contractor's fabrication team shall be involved with the design development.

Materials delivered to the Sub-contractor's works shall have supporting documentation verifying compliance with this Contract Document.

Methods used in the fabrication shall be selected to achieve the specified performance. Methods used shall be based upon the use of suitable equipment and experienced operatives.

The assembly of components shall be carried out in a factory environment to the extent that it is possible to do this.

Components shall be checked by the Sub-contractor for compliance with his drawings and method statements prior to assembly in accordance with his quality procedures.

All changes shall be recorded on a master set of documents and shall be included in the Operations and Maintenance manual as a record of the as-built works.

### 10.2 Assembly

#### 10.2.1 Framework

Grinding, cutting and shaping of metals shall be carried out using tools which will not contaminate them with particles which could stain or corrode them.

Joints shall be accurately formed without lipping or offsets in visible surfaces unless designed otherwise. Other joints shall be rigidly secured to prevent all but designed movement, unless shown otherwise.

Metals shall be welded in accordance with the relevant standards using methods to avoid distortion. The type, size and spacing of welds shall be shown on drawings and reviewed by the Architect, prior to fabrication.

Welds shall be fully bonded throughout their length without holes, inclusions, cracks or porosity so that the long term performance is not compromised and the welds are strong enough for the design requirements. The welds shall be ground smooth and flush with the adjoining surfaces where visible or impinging on other work. Site welding shall not be carried out. The completed welded construction shall be finished to prevent corrosion. All welds shall be cleaned as necessary to ensure the durability of the connections.

### 10.2.2 Mild steel for hot dip galvanising

Mild steel cut or shaped by either flame cutting or Plasma cutting shall be to procedures agreed by the Architect. These procedures shall demonstrate that the surface hardness is less than 270 Hv10. Random inspection of the steel shall be required to ensure that the hardness level is not exceeded. Alternatively, all cut edges shall be surface dressed to remove hardened material.

All punched holes shall be undersized by 1.0mm and be reamed to the finished size.

All radius edges used in changes of section shall be typically 20mm.

All welding procedures and welding procedure specifications shall limit the hardness in the weld, parent metal or heat affected zone to less than 270 Hv 10.

Following pickling operations, the steel shall be held at more than 100 deg C to ensure a uniform temperature, either prior to, or after fluxing.

### 10.2.3 Cutting, drilling and milling

Arc cutting and acetylene gas cutting shall not be used.

### 10.2.4 Pressing and folding

The folding and forming of aluminium sheet shall be carried out over dies or mandrels with the minimum radii recommended by the Aluminium Federation in such publications as The Properties of Aluminium and Its Alloys and The Designer's Guide to rolled Aluminium, Part 2. Trial bends shall be made on scrap material of the alloy, temper and thickness being used to ensure that it is not cracked, torn, unevenly stretched or separated at grain boundaries.

### 10.2.5 Welding of aluminium

Welding procedures shall comply with EN 15614-2.

Welders shall be certified to BS EN ISO 9606-2.

Welding shall comply with BS EN 1011-4.

Welding shall be tested in accordance with BS EN 895, BS EN 1320, BS EN 1321 and BS EN ISO 5173.

All stud welding shall be by the capacitor discharge process on freshly cleaned, degreased and oxide-free base metal shielded with argon gas, 99.95% pure, in accordance with Recommended Practice for Stud Welding by the American Welding Society.

### 10.2.6 Welding of steels

Welding procedures shall comply with EN ISO 15614-1.

Welders shall be certified to BS EN 287.

Welding of carbon manganese steels shall comply with BS EN 1011-2.

Welding of stainless steels shall comply with BS EN 1011-3.

Testing of welds shall comply with the National Structural Steelwork Specification, 5th Edition.

## **10.3 Sealant Application**

### **10.3.1 Sealant application (non-structural)**

Sealant shall be applied to clean surfaces, primed as necessary, in accordance with the manufacturer's recommendations and procedures following adhesion and application tests. Conditions of application and cure shall be conducive to producing satisfactory results and avoid undue stress in uncured material.

Quality control adhesion tests shall be carried out throughout the period of manufacture to monitor the effectiveness of the material and application techniques.

Sealant shall be applied over bond breakers/backer rods, set into the joint to control the depth of sealant and to prevent three-sided adhesion.

Backing rods shall be selected and used strictly in accordance with the sealant manufacturers' recommendations.

### **10.3.2 Sealant application (structural)**

Structural sealant requirements are described elsewhere in this Contract Document.

## **10.4 Bronze**

### **10.4.1 General**

Copper and copper alloy work will be installed in accordance with the requirements of CP 143: Part 12, and the Copper Development Association guidelines.

### **10.4.2 Cutting**

#### **10.4.2.1 Process**

All materials will be cut by automatic or semi-automatic processes

#### **10.4.2.2 Flame cutting**

Hand flame cutting will not be used.

### **10.4.3 Dressing**

#### **10.4.3.1 Dressing**

The appointed Façade Contractor will dress the edges of plate cut by thermal cutting techniques to remove any material with an adverse microstructure, scale, and slag and surface irregularities.

#### **10.4.3.2 Grinding**

The approved Façade Contractor will remove burrs, sharp and ragged edges by grinding. He will ensure that appropriate grinding materials are used for each respective material type and will take appropriate measures to prevent cross-contamination from other materials.

### **10.4.4 Straightening**

#### **10.4.4.1 Material properties**

No straightening is to be carried out which may result in cracking or material properties that do not conform to the minimum requirements for the as-supplied materials.

#### **10.4.4.2 Procedures**

The appointed Façade Contractor will provide straightening procedures for review by the Architect before commencing the work.

### **10.4.5 Heating**

#### **10.4.5.1 Material properties**

No heating is permitted which may result in material properties that do not conform to the minimum requirements for the as-supplied materials.

#### **10.4.5.2 Procedures**

The appointed Façade Contractor is expected to provide procedures for heating for review by the Architect before commencing the works.

### **10.4.6 Welding**

Unless agreed otherwise the welding of copper and copper alloys will not be permitted.

### **10.4.7 Tolerances**

At the time of tender the appointed Façade Contractor will provide details of the manufacturing tolerances to which they will work, and how these will be monitored during production.

## 10.5 Glass

### 10.5.1 Preparing panes of glass

Annealed glass panes, including those for double glazed units, shall be cut to size in the factory by scoring to produce clean cut edges. These edges shall not be dressed, worked or otherwise modified.

The edges of all panes and double glazed units shall be inspected during processing and before and after installation in their frames. Any edges with defects likely to initiate cracks in service including feather (shark's teeth) or whisker (serration's) deeper than half the thickness' of the pane, vents (flake chips) greater than 6mm across, or impact damage shall not be incorporated in the Works.

### 10.5.2 Insulating glass units

Insulating glass units shall comply with BS EN 1279. The visual quality of the installed units shall be assessed in accordance with requirements given elsewhere in this specification.

### 10.5.3 Preparing insulating glass units

The spacers separating the panes of glass shall have adequate rigidity for their purpose. They shall be specifically designed to accommodate the seal and contain desiccant, allowing both to operate at maximum efficiency. Joints in edge spacers shall be limited to a minimum practicable number in any one assembly. Spacer pieces should not be less than 200mm in length. Joints shall not be allowed within 50mm of any corner.

The units shall have a mechanically applied primary polyisobutylene seal, located between glass and spacer providing a vapour-proof barrier, which shall be continuous and of a minimum width of 3mm, and a secondary two part silicone seal extending around the perimeter of the double glazed unit.

Polysulphide secondary seals shall only be used in fully framed systems.

Glass coatings shall overlap the primary seal with no uncoated gap.

Any breather tubes used for transportation shall be nipped closed.

Where argon filled units are proposed, the supplier of the units shall provide written evidence that the units have undergone accelerated ageing tests in accordance with BS EN 1279 -3.

The units shall not employ edge tape. The edges shall be finished in a way which permits visual inspection of the glass edge, edge seals and the positions of the metal spacers.

Identification marks and /or manufacturers' logos shall be sized as small as possible. They shall be located in the bottom left hand corner of the double glazed units when viewed from the inside to the outside.

Pressure release tubes shall be located so that they point downwards in the finished works.

### 10.5.4 Preparing SSG units

Extruded gaskets shall be used in visibly sensitive locations. There shall be no bubbles in the sealant from all viewing angles.

## 10.6 Gaskets

Gaskets jointed by bonding with adhesive are not accepted as a substitute for moulded frame gaskets. In the event that adhesive bonded joints are required to be formed, this shall be brought to the attention of the Architect for review before such work is carried out.

All gasket frames should be normally manufactured to a small but predetermined oversize tolerance, to ensure that when seated into position, the lineal lengths and corners of the gaskets are in slight compression.

Gaskets shall be installed in accordance with the manufacturer's recommendations and utilising the correct tools.

## 10.7 Membranes

Membranes shall be fixed in accordance with manufacturers recommendations based on the application required.

## 10.8 Glass

### 10.8.1 Annealed glass cutting sizes

Tolerances on cut annealed glass will be in accordance with BS EN 572-2.

### 10.8.2 Thermally toughened and heat strengthened glass

After processing, individual pieces of heat treated glass will be within the following tolerances.

|                                         |                                |
|-----------------------------------------|--------------------------------|
| Length and width                        | As required by BS EN 12150     |
| Squareness measured along the diagonals | As required by BS EN 12150     |
| Roller wave                             | 0.15mm between peak and trough |
| Edge dip                                | 0.25mm                         |
| Bow/Local bow                           | As required by BS EN 12150     |
| Length and width                        | As required by BS EN 12150     |

|                                         |                                |
|-----------------------------------------|--------------------------------|
| Squareness measured along the diagonals | As required by BS EN 12150     |
| Roller wave                             | 0.15mm between peak and trough |
| Edge dip                                | 0.25mm                         |
| Bow/Local bow                           | As required by BS EN 12150     |

Optical distortion D in millidiopters is defined as:

$D = 4 \times 10^6 \pi^2 W / L^2$  where W = amplitude (peak to trough) of the roller wave in mm and L = wavelength (peak to peak) of the roller wave in mm.

When assessing the visual effects of roller wave, the optical distortion will not be more than 110 millidiopters.

### 10.8.3 Laminated glass

After laminating, individual pieces of laminated glass will be within the tolerances given in BS EN ISO 12543-5.

### 10.8.4 Insulating glass units

Finished double glazed units shall be within the tolerances given below

|                                                                      |                                                                                                                                                      |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Length and Width                                                     | +0mm, - 2mm                                                                                                                                          |
| Thickness                                                            | + or – 1.0mm                                                                                                                                         |
| Length of diagonals                                                  | 4mm max. difference in length                                                                                                                        |
| Glass alignment along bottom edge.                                   | + or – 0.5mm                                                                                                                                         |
| Glass alignment on other edges                                       | + or – 1.0mm                                                                                                                                         |
| Stripping of coating on surface 2 and or 3 of the double glazed unit | Completely stripped up to nominal value of + or – 1.0mm. Over-stripping max 1.0mm. Edge of coating shall be overlapped by butyl by not more than 2mm |

### 10.8.5 Metal panels

The maximum allowable tolerances for metal panels shall be as follows:

|                                      |                                                                                                                                                                                                                                                                                 |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Length and width                     | $\pm 1.0$ mm up to 120 mm<br>$\pm 1.0$ mm up to 2000 mm<br>$\pm 1.5$ mm up to 4000 mm                                                                                                                                                                                           |
| Depth                                | $\pm 0.5$ mm up to 150 mm deep                                                                                                                                                                                                                                                  |
| Squareness measured on the diagonals | $\pm 0.5$ mm on length of diagonal of sheet and not more than 1.0mm difference in length between the two diagonals for sheet of length and width up to 1200mm, $\pm 1.0$ mm and 1.5mm up to 2000mm, $\pm 2.0$ mm and 2.5mm up to 4000 mm, $\pm 2.5$ mm and 4.0 mm up to 8000 mm |
| Edge straightness                    | $\pm 0.5$ mm over 1.0 m measured from the true line                                                                                                                                                                                                                             |
| Bow/flatness                         | 0.2% of longest panel side up to a maximum of 4.0 mm measured from the true surface                                                                                                                                                                                             |
| Twist                                | One corner not more than 3.0 mm up from the plane containing the other three corners                                                                                                                                                                                            |

### 10.8.6 Louvres

After manufacture, all finished louvre blades within panels shall be within the following tolerances.

|                                                                               |              |
|-------------------------------------------------------------------------------|--------------|
| Length of louvre blade                                                        | $\pm 0.5$ mm |
| Position of louvre blade in plan and elevation                                | $\pm 1.0$ mm |
| Bow of louvre blade                                                           | $\pm 1.0$ mm |
| Max. deviation between each individual louvre blade in elevation and in plane | $\pm 1.0$ mm |

## 10.9 Stone

All cutting finishing and edge labours will be carried out by personnel skilled in the trade, using modern, well maintained equipment.

Stones will be formed to the required shapes without flaws and defects.

Every precaution will be taken to ensure that the stone is not subjected to excessive heat generated by any grinding, shaping, drilling or polishing processes. Suitable lubricants will be used where appropriate to dissipate or reduce the heat generated, and will not cause staining.

Drilling will be undertaken without the use of percussive tools. Only diamond or tungsten tipped bits will be used, together with appropriate cooling methods and drill speeds. All drilling will be carried out in the factory using jigs.

Exposed surfaces and edges will be thoroughly cleaned of all rust stains, iron particles and cutting fluids.

All equipment used will be fit for its intended purpose and be capable of achieving the tolerances prescribed.

### 10.9.1 Stone tolerances

Individual stone elements will be manufactured in accordance with Section 4 of BS EN 1469.

### 10.9.2 Cracks and fissures

All stones will be inspected, particularly at fixing positions and those with minute cracks or natural fissures at these vulnerable places will be brought to the attention of the Architect.

### 10.9.3 Repairs

Except in exceptional circumstances all damaged stonework will be replaced with new stone. In special circumstances, repairs may be necessary, but shall be limited to 20mm in size. However, no on-site or off-site repair of damaged stonework will be carried out without the prior written agreement from the Architect. Before agreeing to repair work, trials will be carried out.

### 10.9.4 Visual inspection

The Façade Contractor will make a standing arrangement for the Architect to inspect at regular intervals any stone element he chooses at the following stages.

- a) After gang sawing and prior to further processing.
- b) After initial processing and cutting to size.

c) After edge labours and surface finishing.

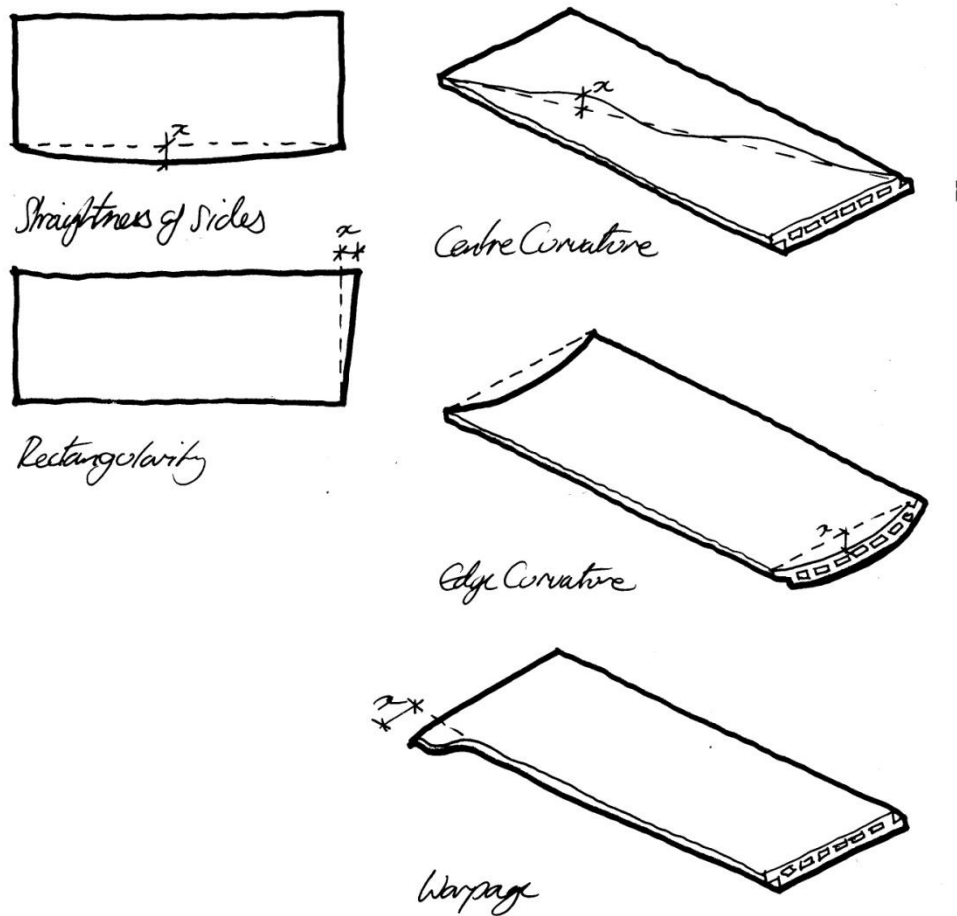
## 10.10 Extruded glazed terracotta

### 10.10.1 General

Panels will be free from cracks, fissures, chips, discoloration, plug marks or any other defects which may adversely affect strength, durability or appearance. Tolerances (non-accumulative)

Tolerances will be measured in accordance with BS EN ISO 10545-2.

- Length (in direction of extrusion):  $\pm 1$  mm;
- Width:  $\pm 2$  mm;
- Thickness:  $\pm 1$  mm;
- Straightness of sides:  $\pm 0.25\%$  of length of edge measured;
- Rectangularity:  $\pm 1$  mm;
- Centre curvature:  $\pm 0.25\%$  of length;
- Edge curvature:  $\pm 0.25\%$  of length of edge measured;
- Warpage: 1.5 mm;
- Lipping between tiles: maximum lipping will be 3 mm overall, but lipping will not exceed 1.5 mm within a batch of units used on the same area of façade;



### ***Definition of terracotta material tolerances***

## **10.11 Sealant Application**

### **10.11.1 Sealant application (non-structural)**

Sealant shall be applied to clean surfaces, primed as necessary, in accordance with the manufacturer's recommendations and procedures following adhesion and application tests. Conditions of application and cure shall be conducive to producing satisfactory results and avoid undue stress in uncured material.

The Contractor shall provide evidence of that the sealant is non staining by obtaining written confirmation from the sealant supplier.

Quality control adhesion tests shall be carried out throughout the period of manufacture to monitor the effectiveness of the material and application techniques.

Sealant shall be applied over bond breakers/backer rods, set into the joint to control the depth of sealant and to prevent three-sided adhesion.

Backing rods shall be selected and used strictly in accordance with the sealant manufacturers' recommendations.

## 11 Installation

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### 11.1 General

As required by the Schedule of Submissions, the Sub-contractor shall prepare method statements and shall submit them to the Architect for review. The review procedure is described in the Contract Documents. The Works shall be installed in accordance with the Sub-contractor's drawings and reviewed method statements, modified as necessary to meet the requirements of the Contract Documents.

Methods used in the installation of the Works shall be selected to achieve the specified performance. The equipment and operatives employed in all stages of installation shall be appropriate to the methods adopted.

Method statements shall set out the Sub-contractor's procedures including programme, sequence of installation, temporary works, dealing with adverse weather conditions, tools and plant, ancillary materials, labour, protection, co-ordination with other elements of work, and regular inspections.

Installation drawings, method statements and quality control procedures shall be supplied to the site installation team. The Sub-contractor shall co-ordinate between the site installation team and the works suppliers and his own designers and site staff so that all those parties are made aware of the important items to be checked during installation.

The method statements and procedures shall detail all the elements used to construct the Works and shall clearly describe all operations and how all CDM requirements are incorporated into the installation process.

No modification or adaptation of the agreed method statements shall be undertaken without the prior agreement of the Architect. All changes to drawings and method statements shall be recorded on a master set of documents and shall be included in the Operations and Maintenance manual as a record of the as-built works.

### 11.2 Site work

#### 11.2.1 General installation requirements

The Sub-contractor shall employ only competent experienced installers to carry out the Works. He shall provide and maintain records to demonstrate his employees experience and all related training.

The Works shall be installed to the dimensions indicated with the permissible deviations estimated by the Sub-contractor and marked on his drawings. These permissible deviations shall be confirmed to the Architect by the Sub-contractor during his detailed design development stage and be appropriate for the materials and methods employed to achieve the project requirements for appearance and fit as specified in the Contract Documents.

As installation proceeds, the Sub-contractor shall check the work and shall record all non-compliance. Remedial proposals shall be reviewed and agreed with the Architect as soon as possible after the problem has been identified.

The Sub-contractor shall not cut, drill or otherwise alter any work without review by the Architect.

### 11.2.2 General appearance

The finished works shall be constructed to the specified tolerances. The same materials and finishes shall be correctly matched to provide a uniform appearance without visibly unacceptable blemishes or variations in colour or texture when viewed from inside and outside in all weather conditions during the day. There shall be no variations in colour or appearance in vision areas due to light reflectance or light transmission when viewed from inside or outside, in all weather conditions during the day and night.

### 11.2.3 On-site control samples

The standard of workmanship for the Works shall be to the standard established by the reviewed on-site quality control samples. Until the Architect has accepted the control samples, the Sub-contractor proceeds with further installation work at his own risk. Upon acceptance the on-site control samples shall remain as part of the Works. The Architect shall reject all workmanship that falls below the accepted standard.

## 11.3 Erection tolerances

### 11.3.1 Windows and doors

### 11.3.2 Tolerances

Glazing will be installed to the following permitted deviations:

|              |                                                                                                           |
|--------------|-----------------------------------------------------------------------------------------------------------|
| Line         | ± 2 mm in any one storey height, or structural bay width;                                                 |
| Level        | ± 2 mm in any one structural bay;                                                                         |
| Plumb        | ± 2 mm in any one structural bay;                                                                         |
| Plane        | ± 2 mm in any one storey height or structural bay width;                                                  |
| Intersection | ± 2 mm in alignment in any direction between any two adjoining panels, at a 4-way intersection of panels; |

### 11.3.3 Hand-set stone

#### 11.3.3.1 Tolerance

Hand-set stone will be erected to the more onerous of the following permitted deviations:

|              |                                                                                                                                                          |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Line         | $\pm 4$ mm in any one storey height, or structural bay width (but no individual stone within $\pm 1$ mm);                                                |
| Level        | $\pm 5$ mm in any one structural bay (but no individual stone within $\pm 1$ mm);                                                                        |
| Plumb        | $\pm 4$ mm in any one structural bay (but no individual stone within $\pm 1$ mm);                                                                        |
| Plane        | $\pm 5$ mm in any one storey height or structural bay width (but no individual stone within $\pm 1$ mm);                                                 |
| Intersection | $\pm 4$ mm in alignment in any direction between any two adjoining panels, at a 4-way intersection of panels (but no individual stone within $\pm 1$ mm) |

### 11.3.4 Mortars

Generally, site practices shall be as described in BS 8000 Part 3. This is the minimum workmanship standard required and industry good practice shall prevail. These site practices shall be strictly adhered to. The Trade Contractor shall ensure that all mortar joints are constructed under the supervision and guidance of an experienced site foreman.

The Contractor shall note that lime is a highly caustic material that requires handling with care, and with the appropriate protective clothing.

Prior to installing mortars, all dust and loose material shall be removed from the stone surfaces. Stone edges shall be damp (but not wet) prior to installation to control suction of water out of the mortar.

Before use lime mortars shall be thoroughly worked into a plastic state that they will retain their shape when moulded in the hand and will cling to the underside of a trowel or hawk. All mortars shall be installed to the full depth of the joint and shall be thoroughly compacted with a thin tool, pointing key or thin wooden tamper. Any spillage shall be washed off immediately with clean water. The joints shall be slightly overfilled and then worked back to a flush profile and finished as it hardens up. Finishing shall involve lightly scraping off the mortar from the overfilled joints to leave an open textured flush surface.

#### 11.3.4.1 Protection and Curing

Finished mortar joints shall be protected from rapid drying due to the sun, wind or artificial heat or from the effects of frost to allow proper curing

### **11.3.5 Structural aluminium work**

Structural aluminium shall be erected in accordance with the tolerances given in BS 8118 Part 2.

## **11.4 Aluminium curtain walls**

The Works shall be installed to the permitted deviations given in sections 7.4.5 and 7.4.6 of part 7 of The Centre for Window and Cladding Technology's Standard for systemised building envelopes.

## **11.5 Bronze elements**

### **11.5.1 Identification**

#### **11.5.1.1 Marking**

The appointed Façade Contractor will mark and document all materials to ensure that they are used as specified.

#### **11.5.1.2 Location of marks**

Piece markings are to be made in positions which are not masked by other material after fabrication and erection. Markings will not be visible on surfaces exposed in the finished Works. The location of piece markings will be submitted to the Architect for approval.

#### **11.5.1.3 Hard stamping**

No copper and copper alloy will be hard stamped.

### **11.5.2 Storage**

#### **11.5.2.1 Holding areas**

The appointed Façade Contractor will store all bronze materials in separate, dry internal holding areas. These areas will be free of chemicals that may be detrimental to the bronze and/or its finishes. The Façade Contractor will ensure that the bronze materials are kept clean.

#### **11.5.2.2 Support**

The bronze materials will be adequately supported clear of the ground in a manner that will not induce stresses or permanent deformations that may be detrimental. Individual piece markings are to be visible when members are stacked.

### **11.5.3 Handling and transportation**

#### **11.5.3.1 General**

The appointed Façade Contractor is expected to plan and carry out bundling, packing, handling and transportation in a manner designed to prevent damage to neither the bonze materials and/or their finishes.

#### **11.5.3.2 Protection**

All assemblies and cladding sections will be adequately protected at all stages of delivery and installation. The appointed Façade Contractor will be responsible for the protection of all installed works such that nothing touches the claddings.

#### **11.5.3.3 Cleaning**

All surfaces will be cleaned and free of debris, dirt and other foreign material.

### **11.6 Visual acceptance criteria for installed glass**

Installed glass will be assessed by using the guideline prepared by the Technical Advisory Board of the Institute of Glazing Trade for Glazing Technology and Window Manufacture, Hadamar (IGH, Hadamar) and the Federal Association for Wholesale Architectural Glazing, Insulating Glass Manufacture, High Performance Glazing, and Troisdorf dated June 2009.

### **11.7 Visual acceptance criteria for installed glass with ceramic frit**

The guideline prepared by the Federal Association for Wholesale Architectural Glazing, Insulating Glass Manufacture, High Performance Glazing and the Professional Association for Architectural Glazing, 2002, titled “Guideline for Visual Assessment of the Visible Quality of Enamelled and Screen-printed Glass” will be used to assess the visual quality of fritted glass installed in the Works.

### **11.8 Extruded glazed terracotta**

#### **11.8.1 General**

Terracotta units will be blended to provide an even distribution of tone and colour across the façade within the agreed ranges of section 9.5.

On-site cutting of the terracotta panels will be avoided unless absolutely necessary. Method statements and sample cuts are expected to be issued by the appointed Facade Contractor and these will be approved by the architect prior to undertaking any cutting.

The terracotta cladding will be set out to match the appearance of the accepted visual mock-ups, as defined in section 7.1.2.

The terracotta cladding will be protected against damage and disfigurement during the course of the works, particularly arrises and corners.

Avoid wiping mortar residue onto the glazed surfaces. Use a wooden tool to remove any clumps of mortar from the glazed surfaces. To clean down after mortar has cured wash with clean running water and a rag or soft bristle brush, repeatedly rinsing out in water.

A detergent type cleaning agent may be used for stubborn areas. Tiles will be wetted prior to application. A final rinse with clean running water will be done immediately. If there is still a milky film on the glazed surface, the area will be washed with a very dilute acid-based cleaner accepted by the manufacturer, then rinsed. This should remove the residue. (Trial and error may be needed to determine an effective mix.)

Cleaners containing hydrofluoric acid will not be used.

The use of jet washing or power washing will not be permitted under any circumstances.

### 11.8.2 Accuracy of Erection – Glazed terracotta panels

The Works will be installed to the following permitted deviations:

|                                                                                |                                                                                                                                             |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Line (straightness on plan)                                                    | ± 3.0mm in any one storey height, or structural bay width                                                                                   |
| Level (horizontality on elevation)                                             | ± 3.0mm in any one structural bay;                                                                                                          |
| Plumb (verticality in vertical section)                                        | ± 3.0mm in any one storey height;                                                                                                           |
| Plane (flatness in all directions planar to face)                              | ± 3.0mm in any one storey height, or structural bay                                                                                         |
| Intersection (flatness in all directions planar to face across unit junctions) | ± 3.0mm in alignment in any direction in the plane of the wall, between any two adjoining cladding units, at a 4-way intersection of panels |

### 11.8.3 Material Protection and storage

Terracotta is expected to be protected in such a manner that will prevent damage, staining, uneven weathering or degradation under normal conditions of handling and storage. Particular attention will be given to the protection of edges, projecting features, corners and other vulnerable areas. The terracotta will be packaged so that delivery in accurate sequence for site fixing is possible. Where undercover storage is not possible, it is essential that protection against frost and staining is provided.

All coated surfaces vulnerable to damage during handling and installation or by subsequent site operations will be fully protected for the duration of the works.

Protective coverings are expected to be resistant to all weathers and be removable from areas inaccessible after installation. They will be partially removable and replaceable for access to fixing points during installation and/or subsequent site operations.

Any protective tapes used in direct contact with the coating will be a low tack, self-adhesive type in white or any colour lighter than the coating to be covered. It will be applied and removed in accordance with the recommendations of the glazed terracotta manufacturer.

## 11.9 Louvre blades

After installation, tolerances on individual louvre blades shall not exceed the following values.

|                                                                               |                    |
|-------------------------------------------------------------------------------|--------------------|
| Length of louvre blade                                                        | $\pm 0.5\text{mm}$ |
| Position of louvre blade in plan and elevation                                | $\pm 1.0\text{mm}$ |
| Bow of louvre blade                                                           | $\pm 1.0\text{mm}$ |
| Max. deviation between each individual louvre blade in elevation and in plane | $\pm 1.0\text{mm}$ |

## 11.10 Fixings

Fixings including primary fixings to the building structure shall be installed in accordance with the fixing manufacturer's recommendations and required procedures, where necessary following site trials and tests. The details of primary fixings including the following shall be checked and recorded:

- Type of proprietary fixings, their size and location
- Embedment, spacing and edge distances
- Torque levels
- Locking washers or nuts and engagement of serrated washers
- Isolation of dissimilar metals and of mill finish aluminium from cementitious material in damp environment.
- Whether the fixing was tested or not. If tested, results of test.
- For all bolt assemblies the strength grade combination of bolts/nuts/washers shall be as prescribed or recommended in the relevant British Standard.

The Sub-contractor shall secure all nuts used in connections to prevent loosening during the life of the Building.

The length of bolt shall be such that at least one clear thread shows above the nut after tightening, and at least one thread plus the thread run out shall be clear between the nut and the unthreaded shank of the bolt.

Where site drilling, cutting, grinding etc is carried out all swarf shall be removed and all surfaces left clean.

## **11.11 Cast-in Channels**

### **11.11.1 General**

Cast-in channels in the concrete structure required for cladding attachment, shall be demonstrated to have sufficient strength and shall be adequately embedded to ensure that the loads applied to them are sufficiently distributed to avoid over-stressing the concrete. The Sub-contractor shall provide cast-in channels and their locations to the concreting Sub-contractor. The concreting Sub-contractor shall co-ordinate these locations with the layout of the steel reinforcement in the concrete and shall install them.

Materials for the cast-in channels shall comply with the requirements of this specification.

### **11.11.2 Protection of cast-in channels**

The concreting Sub-contractor shall ensure that the cast-in channels do not get blocked with concrete, concrete slurry, or any other material that prevents the cladding fixings from being installed.

### **11.11.3 Tolerance on placement**

Cast-in channels shall have a permitted deviation of  $\pm 15\text{mm}$  from the required position. The minimum distance from the edge of any cast-in channels and a concrete edge shall not be less than 100mm.

After installation, the exact position and condition of the cast-in channels shall be checked and verified by the Sub-contractor. Any discrepancy or in-accuracy shall be reported to the Architect immediately.

### **11.11.4 Remedial work**

Where remedial action is required to rectify out-of-tolerance or damaged cast-in channels, the Sub-contractor shall submit proposals for review by the Architect as soon as possible after the discrepancies have been established and confirmed. The Architect prior to any remedial work being carried out on site shall agree these proposals.

The Sub-contractor shall verify the adequacy of all remedial work by means of load testing. The testing regime shall be developed in conjunction with the manufacturer of the anchors proposed and the Architect, and shall be agreed by the Architect and manufacturer

prior to any remedial work being carried out on site. Testing methods shall be those described the Construction Fixing Association Guidance Note: “Procedure for site testing construction fixings.”

## 11.12 Gaskets

Gaskets shall be installed as described elsewhere in this Contract Document.

## 11.13 Insulation

Insulation shall be installed as shown on the Sub-contractor's drawings in accordance with manufacturer's recommendations. It shall be held in place by non-ferrous fixings mechanically attached to the supporting components. Installations relying upon adhesives only shall not be permitted. Where mechanical fixings pass through waterproof membranes or vapour barriers, they shall be properly sealed to the waterproof membranes or vapour barriers.

The method of attachment to the supporting components shall be selected to eliminate the risk of bulging, sagging, delamination or detachment of the insulation. It shall be installed so that there are no gaps or lips between individual pieces of insulation or between insulation and any penetrations through the insulation.

## 11.14 Membranes

Membranes used for flashing, waterproofing and vapour control shall be continuous and shall be fixed mechanically and with adhesive to the backing components and completely sealed. They shall be formed into shapes compatible with the building interfaces. They shall be sealed at all joins, and fixing penetrations.

Where the membrane manufacturer confirms that the service life of the complete membrane system without mechanical fixing and clamping is equivalent to the specified service for the Works, then mechanical fixing and clamping of membranes may be replaced by adhesive bonding.

The closures of junctions at interfaces shall be formed of membrane, on its own for narrow gaps and in combination with aluminium alloy sheet for wider gaps where support is required to prevent damage under design wind load.

Joins between the various pieces of membrane shall be made by gluing and clamping them together where they are supported by a firm substrate and by gluing, vulcanising, solvent welding or heat welding them together where they are unsupported along the line of the joint.

Membranes forming aprons and collars around windows and glazed walls shall be made completely continuous by gluing, vulcanising, solvent welding or heat welding the joints between their various segments.

Membranes shall be attached to the supporting components by sealing and clamping them to the supporting components. The clamps shall be sufficiently stiff and shall be fixed with fasteners sufficiently close together to maintain a complete seal and sufficiently strong to resist any design wind load on the barrier.

Damaged flashings, waterproofing or vapour control layers shall be rejected and shall be completely replaced.

## 12 Maintenance

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### 12.1 Operations and maintenance manual

The Sub-contractor shall produce an Operations and Maintenance manual for the Works. Two copies of the manual will be required. A draft copy of the proposed manual shall be submitted to the Architect not less than six weeks before the programmed date for Completion. The manual shall be developed in parallel with the design and shall include, but not be limited to, the following information:

- a) The name, address and telephone number of each firm and/or Contractor involved in the supply or fabrication of materials, components, assemblies and finishes.
- b) A clear and concise description of the construction used to form the various elements within the Works. It shall be set out in chapters dealing with each element in turn. Its contents shall be comprehensively tabulated and its text cross-referenced. It shall be illustrated with sketches and Trade Literature and shall refer to the 'as-built' drawings.
- c) Copies of material, components and finishes certification and test reports and photographs as required by the Contract Documents.
- d) A method statement based on the access strategy shown on the Contract Documents showing the means of access to all parts of the Works with recommended safe loading.
- e) A method statement covering the procedures for replacement of damaged or otherwise defective materials or components, and materials and components that have a design life less than the design life of the Works and will therefore require replacement during its life.
- f) Quality management record sheets including records of repairs to coatings.
- g) Recommendations and procedures for inspections, routine maintenance, lubrication, cleaning, suitable cleaning agents and painting.
- h) Guidance and recommendations for effective and efficient operation of solar and thermal control features on the building façade with regards to their energy and environmental objectives.
- i) The Method Statement and Disassembly Plan as defined in Clause 2.3.1 of this Specification.
- j) A full set of as-built construction drawings, updated to include any changes made up to the time of completion.
- k) The terms and conditions of any Warranties. Note that all Warranties for finishes shall be valid for periods of one year between cleaning.

Following hand over of the Building at Completion, the Sub-contractor shall allow for providing an experienced member of staff to familiarise the Facilities Manager of the

Building with the use of, and information contained within, the Operations and Maintenance manual.

## 12.2 Bronze

Comprehensive instructions for the maintenance of the bronze finishes shall be provided by the supplier.

The recommendations will include, but not be limited to the following:

- Frequency of washing;
- Solutions/detergents to be used and recommended dilution levels;
- Type of cleaning materials;
- Re-waxing and/or re-lacquering and the associate process details;

The appointed Façade Contractor is expected to take responsibility for ensuring all finishing works conforms to statutory Public Health and Safety regulations.

## 12.3 Extruded Glazed Terracotta Cladding

Comprehensive instructions for the maintenance of the glazed terracotta shall be provided by the Contractor. The appointed Façade Contractor is expected to ensure that recommendations have been endorsed by the glaze manufacturer and that, when applied as specified, they will fulfil the requirements of the Warranty.

The recommendations will include, but not be limited to the following:

- Frequency and recommendations for repointing;
- Frequency of washing;
- Solutions/detergents to be used and recommended dilution levels;
- Type of cleaning materials;
- Recommendations (if any) for deep cleaning processes (including solutions/compounds etc.) and their frequency;
- Cleaning: specify acceptable detergents etc. and methods to be used.

## A1 APPENDIX A

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# PLP Architecture

## Appendix A

### Window performance

Job number 226066/00

02 June 2016

# Introduction

This Appendix shows the thermal performance and glass build-up assumptions for the different window types of the 60 Curzon Street project in London.

## Reference documents

The window types and corresponding dimensions have been taken from the Stage 4.1 Issue package from PLP.

## Overall U-value

The overall U-value has been calculated in accordance with BS EN ISO 10077-1:2006 taking into account for the following components:

- U-value of the frame;
- Centre pane U-value of the glazing system;
- Linear thermal transmittance of the stainless steel spacer in the glazing system;

The overall U-value has been assessed considering Schueco aluminium frame AWS 90 BS.SI (block system).

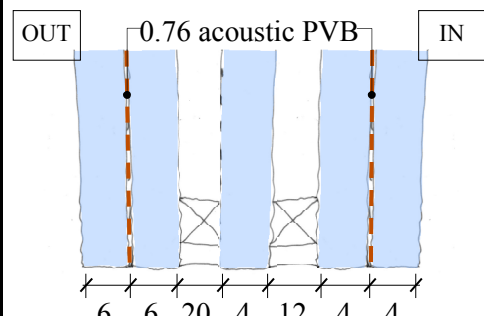
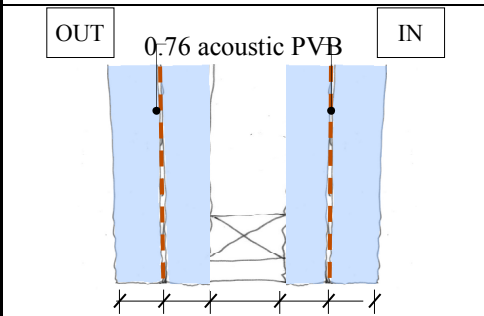
For the Triple Glazing Unit (TGU) a centre pane U-value of 0.6 W/m<sup>2</sup>K has been assumed.

For the Double Glazing Unit (DGU) a centre pane U-value of 1.1 W/m<sup>2</sup>K has been assumed.

## Weight of opening vent

The weight of the opening vent has been calculated considering the overall dimensions and assuming weight of the glazing unit as shown in the table on the top left. Therefore, with this assumption, the weight due to the aluminium frame has not been included as already covered by the glazing unit.

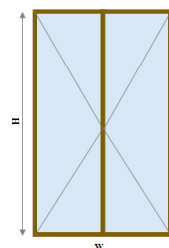
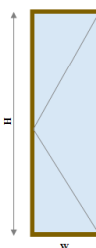
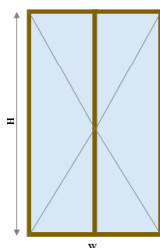
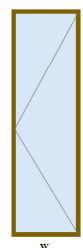
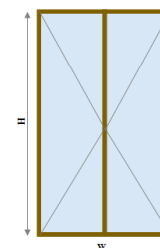
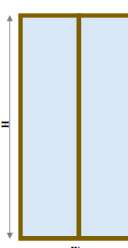
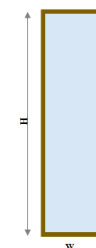
# Window Performance 1/2 Summary Table

|                                                                                   |                         |                                |                             |
|-----------------------------------------------------------------------------------|-------------------------|--------------------------------|-----------------------------|
|  | Total thickness<br>(mm) | Weight<br>(kg/m <sup>2</sup> ) | Ucp<br>(W/m <sup>2</sup> K) |
|                                                                                   | 57.52                   | 60                             | 0.6                         |
|  | 41.52                   | 50                             | 1.1                         |

 Openings that require additional provisions (e.g. stiffeners) due to opening height > 2.2m.

 Openings where weight is greater than 170Kg.

 Openings belonging to Restaurant (DGU).

|                                  |               | BESPOKE THRESHOLD                                                                   |      |      |      |       |      |      |      |      |       |                                                                                       |      |      |      |      |      |      |      | CURVED                                                                                |        | FIXED  |        |                                                                                       |        |                                                                                       |        |                                                                                       |       |                                                                                       |      |      |      |      |       |   |
|----------------------------------|---------------|-------------------------------------------------------------------------------------|------|------|------|-------|------|------|------|------|-------|---------------------------------------------------------------------------------------|------|------|------|------|------|------|------|---------------------------------------------------------------------------------------|--------|--------|--------|---------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------|------|------|------|------|-------|---|
| Opening type                     |               | DOUBLE OPENING                                                                      |      |      |      |       |      |      |      |      |       | SINGLE OPENING                                                                        |      |      |      |      |      |      |      | DOUBLE OPENING                                                                        |        |        |        | SINGLE OPENING                                                                        |        | DOUBLE OPENING                                                                        |        | DOUBLE OPENING                                                                        |       | SINGLE OPENING                                                                        |      |      |      |      |       |   |
| Image                            |               |  |      |      |      |       |      |      |      |      |       |  |      |      |      |      |      |      |      |  |        |        |        |  |        |  |        |  |       |  |      |      |      |      |       |   |
| ID Opening type                  |               | DO-A                                                                                | DO-B | DO-C | DO-D | DO-E  | DO-F | DO-G | DO-I | DO-J | DO-K  | DO-L                                                                                  | SO-A | SO-B | SO-C | SO-D | SO-H | SO-I | SO-J | SO-L                                                                                  | DO-D-B | DO-E-B | DO-F-B | DO-G-B                                                                                | DO-J-B | SO-G-B                                                                                | SO-L-B | SO-N-B                                                                                | CRV_A | CRV_B                                                                                 | DF-D | DF-G | DF-J | SF-E | SF-G  |   |
| H (m)                            |               | 1.65                                                                                | 1.98 | 2.31 | 2.64 | 2.805 | 2.64 | 2.1  | 2.31 | 2.64 | 3.136 | 2.97                                                                                  | 1.95 | 1.95 | 2.61 | 2.78 | 1.65 | 1.98 | 2.15 | 2.61                                                                                  | 2.64   | 2.805  | 2.64   | 2.1                                                                                   | 2.64   | 2.475                                                                                 | 2.61   | 2.97                                                                                  | 1.65  | 1.98                                                                                  | 2.64 | 2.1  | 2.64 | 2.1  | 2.475 |   |
| W (m)                            |               | 1.2                                                                                 | 1.2  | 1.2  | 1.2  | 1.2   | 1.5  | 1.54 | 1.8  | 1.8  | 1.8   | 1.65                                                                                  | 0.44 | 0.60 | 0.60 | 0.60 | 0.75 | 0.80 | 0.80 | 0.80                                                                                  | 1.2    | 1.2    | 1.5    | 1.54                                                                                  | 1.8    | 0.725                                                                                 | 0.80   | 1.00                                                                                  | 0.95  | 0.95                                                                                  | 1.2  | 1.54 | 1.8  | 0.68 | 0.725 |   |
| Area (m²)                        |               | 1.98                                                                                | 2.38 | 2.77 | 3.17 | 3.37  | 3.96 | 3.23 | 4.16 | 4.75 | 5.64  | 4.90                                                                                  | 0.86 | 1.17 | 1.57 | 1.67 | 1.24 | 1.58 | 1.72 | 2.09                                                                                  | 3.17   | 3.37   | 3.96   | 3.23                                                                                  | 4.75   | 1.79                                                                                  | 2.09   | 2.97                                                                                  | 1.57  | 1.88                                                                                  | 3.17 | 3.23 | 4.75 | 1.43 | 1.79  |   |
|                                  | AWS 90 BS.SI+ |                                                                                     |      |      |      |       |      |      |      |      |       |                                                                                       |      |      |      |      |      |      |      |                                                                                       |        |        |        |                                                                                       |        |                                                                                       |        |                                                                                       |       |                                                                                       |      |      |      |      |       |   |
| Glass weight single leaf (kg/m2) | TGU           | 59                                                                                  | 71   | 83   | 95   | 101   | 119  | 97   |      | 143  | 169   | 147                                                                                   | 51   | 70   | 94   | 100  | 74   | 95   | 103  | 125                                                                                   | 95     | 101    | 119    | 97                                                                                    | 143    | 108                                                                                   | 125    | 178                                                                                   | 94    | 113                                                                                   | -    | -    | -    | -    | -     |   |
|                                  | DGU           |                                                                                     |      | 69   | 79   |       |      |      | 104  | 119  |       |                                                                                       | 43   |      |      |      |      | 79   |      | 104                                                                                   |        |        |        |                                                                                       |        |                                                                                       | 104    |                                                                                       |       |                                                                                       |      |      |      |      |       |   |
| U (0.60) (W/m2K) - TGU           |               | AWS 90 BS.SI+                                                                       | 1.10 | 1.10 | 1.10 | 1.10  | 1.00 | 1.00 | -    | 0.94 | 0.94  | 0.96                                                                                  | 1.20 | 1.10 | 1.10 | 1.10 | 1.00 | 1.00 | 0.99 | 0.98                                                                                  | 1.20   | 1.20   | 1.10   | 1.10                                                                                  | 0.99   | 1.10                                                                                  | 1.00   | 0.97                                                                                  | 1.30  | 1.30                                                                                  | -    | 0.93 | -    | 0.96 | 0.94  |   |
| U (1.1) (W/m2K) - DGU            |               | AWS 90 BS.SI+                                                                       | -    | -    | 1.30 | 1.30  | -    | -    | -    | 1.30 | 1.30  | -                                                                                     | 1.40 | -    | -    | -    | -    | 1.30 | -    | 1.30                                                                                  | -      | -      | -      | -                                                                                     | -      | -                                                                                     | -      | 1.40                                                                                  | -     | -                                                                                     | -    | 1.40 | -    | 1.30 | -     | - |

# Window Performance 2/2 Summary Table

|  |                         |                                |                                         |
|--|-------------------------|--------------------------------|-----------------------------------------|
|  | Total thickness<br>(mm) | Weight<br>(kg/m <sup>2</sup> ) | U <sub>cp</sub><br>(W/m <sup>2</sup> K) |
|  | 57.52                   | 60                             | 0.6                                     |
|  | 41.52                   | 50                             | 1.1                                     |

- Openings that require additional provisions due to opening height > 2.2m.
- Openings which weight is greater than 170Kg.

|                                  |               | BESPOKE THRESHOLD     |       |       |                       |       |                       |                       |         |                       |         |
|----------------------------------|---------------|-----------------------|-------|-------|-----------------------|-------|-----------------------|-----------------------|---------|-----------------------|---------|
| Opening type                     |               | HYBRID DOUBLE OPENING |       |       | HYBRID TRIPLE OPENING |       | HYBRID SINGLE OPENING | HYBRID DOUBLE OPENING |         | HYBRID TRIPLE OPENING |         |
| Image                            |               |                       |       |       |                       |       |                       |                       |         |                       |         |
| ID Opening type                  |               | HYB-B                 | HYB-C | HYB-D | HYB-E                 | HYB-F | HYB-A-B               | HYB-B-B               | HYB-C-B | HYB-E-B               | HYB-F-B |
| H (m)                            |               | 2.64                  | 2.81  | 1.65  | 2.64                  | 2.81  | 2.48                  | 2.64                  | 2.81    | 2.64                  | 2.81    |
| W (m)                            |               | 2.10                  | 2.10  | 2.10  | 3.62                  | 3.62  | 1.58                  | 2.10                  | 2.10    | 3.62                  | 3.62    |
| Area (m <sup>2</sup> )           |               | 5.54                  | 5.89  | 3.47  | 9.56                  | 10.15 | 3.91                  | 5.54                  | 5.89    | 9.56                  | 10.15   |
| Glass weight single leaf (kg/m2) | AWS 90 BS.SI+ |                       |       |       |                       |       |                       |                       |         |                       |         |
|                                  | TGU           | 91                    | 97    | 57    | 91                    | 97    | 122                   | 91                    | 97      | 91                    | 97      |
|                                  | DGU           | 76                    |       | 47    |                       |       |                       | 76                    |         |                       |         |
| U (0.60) (W/m2K) - TGU           |               | AWS 90 BS.SI+         | 1.10  | 1.10  | 1.10                  | 1.10  | 1.10                  | 1.20                  | 1.20    | 1.20                  | 1.20    |
| U (1.1) (W/m2K) - DGU            |               | AWS 90 BS.SI+         | 1.40  | -     | 1.40                  | -     | -                     | 1.50                  | -       | -                     | -       |